Instructional Strategies for Middle and High School

Bruce E. Larson
and Timothy A. Keiper

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Routledge
Taylor & Francis Group
New York London

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Routledge
Taylor & Francis Group
711 Third Avenue,
New York, NY 10017

Routledge
Taylor & Francis Group
2 Park Square
Milton Park, Abingdon
Oxon OX14 4RN

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Routledge is an imprint of Taylor & Francis Group, an Informa business

International Standard Book Number-10: 0-415-95309-X (Softcover) 0-415-95308-1 (Hardcover)
International Standard Book Number-13: 978-0-415-95309-2 (Softcover) 978-0-415-95308-5 (Hardcover)

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Library of Congress Cataloging-in-Publication Data

Larson, Bruce.
Instructional strategies for middle and high school / Bruce Larson, Timothy Keiper.
p. cm.
Includes index.
ISBN 0-415-95308-1 (hb) -- ISBN 0-415-95309-X (pb)
1. Teaching--Handbooks, manuals, etc. 2. Lesson planning--Handbooks, manuals, etc. 3. Educational tests and measurements--Handbooks, manuals, etc. 4. Classroom management--Handbooks, manuals, etc. I. Keiper, Timothy. II. Title.

LB1025.3.L367 2007
373.1102--dc22 2006022158

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http://www.taylorandfrancis.com

and the Routledge Web site at
http://www.routledge-ny.com

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ACKNOWLEDGMENTS

We acknowledge the important roles many have played in the formation and development of the ideas in this book. Our students and colleagues at Western Washington University, and the many middle school and high school teachers we have the pleasure to work with each year, provided many creative ideas, feedback, and suggestions. Our families have been a continual source of encouragement as we engaged in the writing process. Their encouragement and support provided us with a foundation to get in the proper frame of mind to write. Many reviewers provided feedback on each chapter that challenged us to express our ideas more clearly. The entire editorial team at Routledge has been an excellent source of encouragement and assistance. In particular, the editorial direction of Catherine Bernard shaped our ideas from the start. Angela Chnapko, who took over from Brook Cosby, provided clear feedback that helped focus our writing and revisions.

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INTRODUCTION

Instructional Strategies for Middle and High School is designed to be the primary text for secondary education teacher methods courses. These courses typically examine curriculum development, instructional strategies, classroom management, and student assessment. This book prepares preservice teachers for effectively using seven powerful instructional strategies and provides very clear guidelines for writing unit plans and lesson plans. Since effective classroom management and valid assessment techniques are important parts of student learning, we integrate practical ways to manage the learning environment and assess student learning for each instructional strategy. In short, this book is written to help teachers be successful. We define successful teachers as those who help all students learn important facts, concepts, skills, and attitudes. Successful teachers also enjoy this process. We believe teaching is a wonderful and exciting profession, and we hope the pages of *Instructional Strategies for Middle and High School* help excite and empower the next generation of classroom teachers.

WHY THIS BOOK?

As we wrote this book we were mindful of our students: preservice middle school and high school teachers who are enrolled in instructional methods courses. We call these students *aspiring* teachers, because they aspire to become effective in helping their students learn. We frequently notice, however, that aspiring teachers struggle with selecting instructional strategies, and often the strategies they select do not line up with the curriculum. For example, a teacher will decide to lecture because it is efficient, and not because it is deemed the best strategy for helping students learn a set of key concepts; or a teacher will use a simulation or group project to "add some variety" to the day, and not because these approaches will promote student learning more effectively than others. Similarly, decisions about the assessment of student

learning, classroom management, use of digital technology, and many other common areas of the classroom are difficult for aspiring teachers to make because each decision is modified based on the content being learned and the instructional activities planned for the class session. Aspiring teachers need guidance as they make decisions, and we believe a book that provides this will be a unique and important contribution to teacher education programs. As a result of these thoughts, we decided to write *Instructional Strategies for Middle and High School.*

LINKING INSTRUCTION, ASSESSMENT, AND MANAGEMENT

Over the course of the text we explore seven categories of instructional strategies. Our decision for including these strategies was informed by conversations with many middle school teachers, high school teachers, and principals. All agree that the seven strategies included herein are most commonly used by teachers, are effective for helping students learn, and are needed for all teachers to understand. By listing step-by-step procedures for using these instructional strategies, this text gives aspiring teachers a deep understanding of each strategy and the tools with which to determine how to best use each with a classroom of adolescents. Along with teaching, beginning teachers often report that they struggle with making valid and authentic assessments of their students' learning. This book provides a concise overview of assessment, and offers many concrete examples of assessments to use with *each* of the instructional strategies. For example, valid assessments of what students learned from a role-playing activity will be different from assessments following a classroom discussion. Appropriate assessments are integrated into each of the chapters for the instructional strategies, with contextualized examples given in order to provide tangible ideas for the reader. We provide examples of unit plans, lesson plans, and rubrics that we have gathered from the many teachers who provided input on this book. In addition to exploring assessments specific to each strategy, we also discuss classroom management within the context of each of the teaching strategies. A common worry for aspiring teachers is how to manage a classroom with thirty students and how to discipline students who are disruptive. We have observed that many student teachers will not try student-centered activities for fear of "losing control" of their students. This book addresses concerns such as these by providing specific classroom management ideas for each of the instructional strategies. Most methods texts we have reviewed do not apply management theory or valid assessments to specific strategies of instruction. *Instructional Strategies for Middle and High School* does, because it is impossible to use a strategy effectively without considering unique management concerns and assessments for each strategy.

All of the ideas presented in this book are theory based, and our practical suggestions are built on extensive examinations of current and long-standing research. In writing this text we have made great efforts to focus on the

day-to-day teaching and learning that need to occur in classrooms because we want to provide future teachers with the tools to proactively manage their classrooms and ensure that all students are able to learn. The following ideas directed our development of the contents:

Effective teaching promotes student learning of the course content; teachers matter!

Students bring diverse abilities, interests, and needs to the classroom that the teacher must accommodate.

Teachers need skill with a large and diverse set of instructional strategies to help their students learn.

The content students will learn will help determine the instructional strategies selected by the teacher.

Each instructional strategy has unique assessments, and teachers must closely match the assessment to the strategy if they are to accurately and validly know what their students have learned.

Aspiring teachers need tools for developing effective classroom management skills and student discipline strategies. Teachers often forget that each instructional strategy presents unique classroom management considerations, so part of a teacher education program needs to apply management theories to each instructional strategy.

Teaching requires constant decision making. As such, decisions about class activities, student discipline, classroom management, and student assessment will be better if the teacher engages in a thoughtful decision-making process.

THE STRUCTURE OF THE BOOK

Instructional Strategies for Middle and High School is divided into two sections. Section 1 is titled "Contexts of Teaching and Learning," and section 2 is titled "Instructional Strategies."

Teaching on a Solid Foundation

The four chapters of section 1 explore theories and research about learning and adolescent development. This provides the foundation for the content of the book. Teaching extends beyond pure methods and strategies, and includes areas such as tone of voice, subtexts of curriculum, classroom environment, and classroom management. This first section describes contextual factors that all teachers need to consider when planning, teaching, assessing learning, and reflecting on their teaching, including:

motivating students to learn

diverse characteristics and needs of students

differentiated instruction
the use of digital technology
guiding principles for classroom management and discipline
national, state, and local standards
determining classroom curriculum
domains of learning
assessing student learning

Additionally, section 1 presents reasons for course goals and objectives, the components of a lesson and unit plan, and ideas for appropriate sequencing of content.

Seven Instructional Strategies

The seven chapters in section 2 provide ideas for teaching effectively, and will help a teacher provide instruction that is engaging, varied, and most appropriate for the content being learned. In each chapter we provide guidance about the process of selecting classroom activities that are appropriate and based on current ideas from educational researchers and theorists. Each chapter is devoted to a specific type of instructional strategy to allow an in-depth examination and provide step-by-step procedures for implementing the strategy in the classroom. To accomplish this, we very purposefully explore eight areas for each strategy:

1. an overview of the strategy
2. a description of what research has revealed about the strategy
3. step-by-step procedures for planning and implementing the strategy
4. a realistic exploration of the logistical concerns related to each strategy, including benefits and obstacles when using the strategy
5. classroom management and student motivation issues unique to the strategy
6. appropriate assessment techniques to use with the strategy
7. guidelines for deciding if a strategy is appropriate for the content of a lesson
8. opportunities for the teacher to consider how to use the strategy with the content to be taught

We start with instructional strategies that are teacher centered and move to strategies that are increasingly student centered. As mentioned earlier, *Instructional Strategies for Middle and High School* highlights classroom management and assessment as subsections for *each* instructional strategy. This integrates and contextualizes the management, assessment, and learning theories developed in the first section of the book. To help aspiring teachers understand this further, the book contains many examples from a wide range of middle school and high school classrooms. We have also compiled, in appendix E, all seven

of the instructional strategies described in the book. It serves as a quick point of reference to help you select appropriate strategies in your classroom.

PUTTING IT ALL TOGETHER

Effective middle school and high school teachers select an instructional strategy for a lesson because they think it will help their students meet the day's learning objectives. Consider this example: If students need to learn the history and context around the development and writing of the U.S. Constitution, why might a teacher use a simulation, as opposed to a class discussion, classroom lecture, or an inquiry-based research project? Might these other strategies be more effective? How will the teacher determine what the students have learned? These questions are not easy to answer, and are a continual part of the teaching process. The strategies that a teacher selects for engaging students require a complex series of decisions that include the learning preferences of the students, the content being learned, the setting of the classroom, assessment procedures, and a teacher's comfort level with a particular approach to leading the class. When a teacher selects an instructional strategy without concern for the content or for student learning, all sorts of problems abound. This text provides clear guidelines for helping aspiring teachers engage in this type of thoughtful and effective decision-making process.

Classrooms are comprised of diverse learners, so we want the reader—the future teacher—to consider how each strategy might allow for differentiated learning. Also, we consider ideas for appropriately integrating digital and computer technology into the lessons. Additionally, each chapter has at least one section titled "Making It Work in Your Content Area." In these sections the reader will be asked to determine how to apply to a future subject area the ideas presented within the chapter. This helps in the process of thinking about teaching a specific content area, and further contextualizes the chapter. We have also found this to be a helpful tool for encouraging student teachers to explore what is taught in middle school and high school, and note the differences and similarities with how subjects are taught on the university level. It is our hope that all teachers will establish classrooms that have a positive impact on all their future students' learning.

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CONTEXTS OF TEACHING AND LEARNING

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1

LEARNING, TEACHING, AND MANAGEMENT

As a future teacher you will face many challenging tasks. Knowing your students is one important issue; classroom management is another. Understanding the perspectives and backgrounds of your students and community and knowing the content of the courses you will teach all provide opportunities and challenges. In addition, your students are wrestling with their roles in the world, and trying to understand their growing and developing bodies as they leave childhood and become young adults. The purpose of this chapter is for you, the future teacher of adolescents, to consider many issues that surround student learning. By understanding how adolescents develop and how they perceive the world, you will be able to more effectively help them learn academic content, intellectual thinking skills, and interactive social skills. Throughout this book we will return to many of the ideas introduced in this first chapter that pertain to learning, teaching, and classroom management. Our intention is for this chapter to provide a foundation for thinking about your future students that will allow you to develop an increasingly complex understanding of how to effectively promote your students' learning.

ADOLESCENT DEVELOPMENT AND INSTRUCTION IN THE SECONDARY SCHOOL

When you walk down the halls of any middle school you will be amazed at the vast physical differences of the students. Some sixth grade boys and girls look as though they are still in elementary school, and some eighth grade boys and girls would appear to be high school students. Even students who are the same age develop very differently. All adolescents' bodies are going through dramatic physical changes, but they are not necessarily experiencing them at

the same time. Your students will differ greatly in their biological, psychological/cognitive, and sociocultural development. The exciting news about this is that opportunities abound when teaching adolescents because of their newly developing cognitive skills, social interactions, and awareness of the world. One fruitful approach for thinking about middle school and high school students is to consider three areas of human development:

1. biological
2. psychological and cognitive
3. sociological and cultural

Biological development considers the influence of the body's physical changes on the adolescent. This applies to learning and teaching because physical changes impact adolescent body size and shape, nutritional needs, mood swings, and the many other changes that are occurring in the body. In many cases your future students will be observers of their own bodies, wondering how changes will affect them. Girls and boys develop at different rates within their same sex, but they are also developing noticeably differently between the two sexes. In sixth grade, girls are often the tallest in the class, and this slowly changes as boys experience growth spurts around the 10th grade. Puberty—with all of the hormonal influences that acompany it—occurs for all students eventually. Mood swings are a huge aspect of hormonal secretion. Not only do moods change, they also are magnified to extremes (e.g., high highs and low lows). The fact that students develop at different rates means you will have some students who are much more physically mature than others. The challenge is to realize that physical development is not linked to social maturity, emotional stability, or cognitive ability. For example, a 15-year-old student may look like an adult, but still have the maturity of a child. Physical changes will affect girls differently than boys (a topic we will explore later in this chapter), and physical changes will affect social, emotional, and academic domains of an adolescent. These physical changes will influence the instructional strategies you choose, the ways you manage the learning environment, your relationships with students, and many other elements of teaching.

Psychological and cognitive development considers how the mind impacts the adolescent. As children grow physically they also increase their cognitive abilities. Thinking skills improve as they develop, and adolescents begin to have cognitive abilities that were only starting to emerge in the fourth and fifth grades. For example, students in middle and high school will become increasingly able to think about possibilities, hypotheses, and abstract concepts. Adolescents gain the exciting ability of introspection, or the abilities to analyze and evaluate themselves and others. Specific theories about how adolescents develop cognitively are examined later in this chapter. The important idea to think about is that your future students have very capable cognitive/thinking abilities. You will help your students enhance these abilities by providing opportunities through which to practice using them. At times this presents a paradox for adolescents and teachers alike. Beamon (2001) suggests

several of these contradictions that may be resolved simply with practice in your classroom:

Adolescents may be technologically "savvy," yet lack skills to organize, evaluate and synthesize data.
Adolescents may be used to fast access of information, yet lack motivation to see a task to completion.
Adolescents may be socially active, yet lack the skills for purposeful social interactions.
Adolescents may be independent minded, yet be vulnerable to peer and societal lures.
Adolescents may be intellectually capable, yet be unpracticed in higher cognitive thinking.

Your classroom becomes an ideal place for developing and refining these cognitive abilities.

Sociological and cultural development receive a tremendous amount of attention when thinking about teaching adolescents because they are closely linked to physical, psychological, and cognitive development. As adolescents struggle with their own identity, they experience a series of identity crises (Erikson, 1959; 1968). These crises are closely tied to self-concept, our ideas/thoughts about who we are. As we make decisions about our identity, we question how we love, with whom we have intimacy, whether we have autonomy and initiative, and the effectiveness of our work ethic. Our culture, ethnicity, economic level, social status, athletic ability, academic prowess, and self-esteem are a few of the many aspects that are part of our identity. They each have a role in shaping our sociological development.

Elkind (1998) suggests that the cognitive and physical development of adolescents puts them sociologically in a time of extreme egocentrism. Children demonstrate this as they deal with an "imaginary audience" and a "personal fable." These two phenomena are initiated in early adolescence, and while we learn to manage them they may never leave us completely. Imaginary audience "accounts for the teenager's extreme self-consciousness. Teenagers feel that they are always on stage and that everyone around them is as aware of, and concerned about their appearance and behavior as they are themselves" (Elkind, 1998, p. 40). The idea of a personal fable stems from their thinking that they are "special, different from other people … [it gives them] courage to participate in many necessary but frightening activities" (pp. 43–44). In addition to creating a sense of invulnerability ("I won't be in a car accident"), the personal fable helps convince adolescents that no one has ever loved so deeply, been so excited, or experienced such emotions as they have. As Elkind summarizes, bringing the fable "in line with reality is part of maturing … only in finding out how much we are like other people, paradoxically, can we really discover our true specialty and uniqueness" (p. 45). The impact of sociological and cultural influences on adolescents, and implications for teachers, will be expanded upon in the coming pages of this book.

Learning

Our adult memories of adolescence can be detailed and specific. However, an adult thinking about her adolescence is very different from the thinking we did when we were actually adolescents. Adult memories of overcoming an obstacle, confronting a peer, interacting with a teacher, or studying for an exam are influenced by the fact that we know how the situation ends. Adolescents are in the process of living out these scenarios without this perspective; they therefore may look at them differently than would adults. When teachers consider how to best help students learn in school, their own memories of school should serve as a guide but not the driving force behind decisions. Certainly, teachers have all been apprenticed by the teachers who taught them. In fact, teachers tend to teach in the manner they were taught (Lortie, 1975). So our memories are useful, but not as reliable as we may think.

It is important for you to think about learning, since that is what teachers help students do in school. Many different theories attempt to understand how people learn. One definition of learning that we find helpful for teachers is provided by Ormrod (2006), who posits that learning is a "relatively permanent change, due to experience, either in behavior or in mental representations or associations" (p. 184). Teachers, then, work with their students to help bring about this change. For example, some suggest that memorizing is not "learning" because it is usually not permanent; it takes rehearsal, and memorized information usually fades over time. Others suggest that memorization is one form of learning because the process has brought about change. More important for you, however, is the realization that your opinion about this will impact the type of instructional strategies you select. If a sixth grade science teacher wants her students to learn the habitats of grizzly bears, she will choose classroom activities that are based on her belief about learning.

Behaviorism is a long-established approach for thinking about learning. According to behaviorists, learning occurs when the overt behavior changes. Over 100 years ago the famous behaviorist experiment by Ivan Pavlov involved a person ringing a bell and feeding a dog. After repeating the stimulus of ringing and feeding, the person rang the bell only. The dog had learned to associate the bell with food, and began to lick its lips in anticipation of food. This idea that behavior (salivating) could be influenced by a particular stimulus (the bell) ushered in the notion of behaviorism. B. F. Skinner believed that an organism's interactions with the world (or the environment) conditions behaviors. In other words, people are continually coming into contact with stimuli. At times these stimuli encourage us (e.g., praise when doing a good job), and at times they discourage us (e.g., punishment). Skinner has suggested that these stimuli shape who we are and what we learn.

The idea that behavior indicates learning is unsatisfying for cognitive approaches to learning. Cognitive approaches attempt to address what is happening inside our brains when we are learning. Jean Piaget, Lev Vygotsky, and other cognitive psychologists investigated how our brains functioned, and

not just our behaviors. For example, Piaget believed that the brain becomes increasingly able to learn complex and abstract information as we grow. Adolescence begins in the final years of Piaget's "Concrete Operations Stage." During this stage children organize their thoughts logically and sequentially. Piaget suggests that near age twelve, children engage in "formal operations," where their thought becomes more abstract. They are able to generate abstract propositions, multiple hypotheses, and possible outcomes. Thinking becomes less tied to concrete reality. Vygotsky theorized about a "zone of proximal development" that exists between our current level of understanding something and the next level. To arrive at this next level, we need help. Tutoring, modeling, and guiding all provide examples of how to expand on our current levels of understanding. Current thinking about ideas tied to Vygotsky's initial theory have led to such terms as *scaffolding*, *constructivism*, and *hands-on learning*. These all imply that learning occurs when some additional source provides a support network to promote learning (e.g., in the same way that a scaffold provides support for a building).

The main goal of learning is to promote student achievement. In *How People Learn*, Bransford, Brown, and Cocking (2000) suggest that three findings consistently emerge from "research on learners learning and teachers teaching" (p. 14) to promote student achievement. Below we list these three, and follow each with a brief explanation.

> Students come to the classroom with preconceptions about how the world works. If their initial understanding is not engaged, they may fail to grasp the new concepts and information that are taught, or they may learn them for purpose of a test but revert to their preconceptions outside the classroom. (p. 14)

Students come to classes with a wide range of experiences and prior knowledge. If this knowledge is accurate, then teachers have the opportunity to tap into a wealth of knowledge that comes from these students. (Imagine how a unit on Japanese culture could be enhanced if one of your students had lived in Japan.) Conversely, misconceptions need to be "unlearned" so that students can relearn proper information. (Middle school and high school science students often have a misconception of the moon's orbit in relation to the Earth and the Sun.)

> To develop competence in an area of inquiry, students must: (a) have a deep foundation of factual knowledge, (b) understand facts and ideas in the context of a conceptual framework, and (c) organize knowledge in ways that facilitate retrieval and application. (p. 16)

Researchers have examined the performances and learning of experts and novices in several subject areas. An expert understanding of key concepts facilitates understanding of new knowledge. This allows experts to identify

patterns, and note similarities or differences in a manner that novices cannot. Therefore, teachers should be able to teach key concepts and/or facts in depth. These "defining concepts" are important if students are to learn additional ideas and information, and they are important for helping students learn content on their own.

> A metacognitive approach to instruction can help students learn to take control of their own learning by defining learning goals and monitoring their progress in achieving them. (p. 18)

Metacognition, or "thinking about one's thinking," frequently takes the form of an internal conversation. Adolescents can be taught effective internal dialogues that will promote learning. For example, predicting outcomes, explaining an idea to yourself to improve understanding, noting when you do not understand, referring to background knowledge, and using strategies for studying are all elements of learning to be metacognitive. These are not used by all students, and should be taught by the teacher through modeling and other explicit ways of bringing out the hidden/internal voice for the students to observe (p. 21).

Learning Style Preferences

Litzinger and Osif describe learning styles as "the different ways in which children and adults think and learn" (1992, p. 73). Often teachers engage students in a wide range of learning activities in order to provide a variety of learning opportunities. Given our earlier definition of learning, *learning preferences* does not mean that students will learn material differently; it means that students may prefer one type of instruction over another. For example, some students will prefer to work on inquiry or research tasks, others will prefer to listen to a lecture, and still others may prefer to see examples as they hear new information. If you believe that all students have different learning preferences, then you will quickly realize that you have preferred styles, too. The challenge is to teach with a variety of styles, and not solely using the style you most prefer.

Multiple Intelligences

The idea of more than one form of intelligence was developed at Harvard University by Gardner, whose *multiple intelligence theory* suggests that longstanding ideas about intelligence such as the intelligence quotient, or IQ, are limited. Gardner notes that our society tends to value linguistic skills (speaking, reading, and writing) and logical-mathematical skills (number sense and reasoning). People who are "well-spoken," "logical," and "calculated" are

highly esteemed. This value might overlook students who have intellectual capabilities in nonlanguage or mathematical areas. Actors, designers, musicians, dancers, counselors, and naturalists may not be valued by some in our society because their capabilities are different. Gardner's multiple intelligence theory (1983, 1999) suggests that there are eight different intelligences, and that these account for a broader conception of what we think about human potential.[1] In a recent keynote address to members of the American Educational Research Association, Gardner stated that he is concerned about educators confusing multiple intelligences with learning styles (2003). Where learning styles are preferences that a learner has in how he is taught, intelligences are learning aptitudes, or ways of thinking about the world around him. The eight intelligences proposed are:

Linguistic/verbal intelligence. This involves abilities with spoken and written language, the ability to learn languages, and the capacity to use language to accomplish certain goals. Writers, poets, lawyers, and public speakers are among those that usually have a high linguistic intelligence.

Logical-mathematical intelligence. This consists of the capacity to think logically, carry out mathematical operations, and investigate issues with a systematic/scientific approach. This intelligence includes the ability to detect patterns, reason deductively, and think logically.

Musical intelligence. This involves skill in the performance, composition, and appreciation of musical patterns. It encompasses the capacity to recognize and compose musical pitches, tones, and rhythms.

Bodily kinesthetic intelligence. This gives rise to the potential of using one's whole body or parts of the body to solve problems. It is the ability to coordinate bodily movements. Gardner sees mental and physical activity as related, since the brain tells the body what to do.

Spatial intelligence. This involves the potential to recognize and use the patterns of wide space and more confined areas. The world around one makes sense because of patterns and organizations that others without this intelligence may overlook.

Interpersonal intelligence. This is the capacity to understand the intentions, motivations, and desires of other people. With this intelligence a person works effectively with others. Educators, salespeople, religious and political leaders, and counselors utilize this intelligence.

Intrapersonal intelligence. This entails the capacity to understand one's own feelings and emotions. With this intelligence a person has a useful understanding of herself and is able to use this information to regulate her life.

Naturalist intelligence. This involves abilities for sensing patterns in and making connections to elements in nature. People with a naturalist intelligence may have a strong affinity to the outside world or to animals, and may show unusual interest in such subjects as biology, zoology, botany, geology, meteorology, paleontology, or astronomy.

Gardner does not suggest that teachers teach to all eight intelligences. He does, however, suggest that teachers will have students who look at the world and understand it in terms of one or more intelligences. The principal implication of this is that teachers should provide a variety of activities that focus on different intelligences. Consider how this might impact you in the classroom. If you were to teach eighth graders in a world geography class, you might have your students read about a particular region in the world (linguistic/verbal), look at a map of that area (spatial), and examine demographic information about that region (logical-mathematical). They could then record their impressions of that region in a journal (intrapersonal) and discuss them in groups of four (interpersonal). What Gardner's theory provides is a challenge for the teacher to help students learn content by thinking about it in ways that benefit the students' strengths.

Making It Work in Your Content Area

Along with all of the cognitive developments that occur as adolescents develop is the ability to analyze and evaluate others and themselves. Think of at least one way that you can make use of these emerging thinking skills when teaching your content area. For example, how might a history teacher use students' introspection, analysis, or evaluation abilities when reading a biography of John Adams? How might a chemistry teacher use them during a lab activity?

MOTIVATING STUDENTS TO LEARN

Motivation might best be thought of as a force, or energizer, that helps us accomplish a task. We are motivated by many things, and when we lack motivation we usually accomplish very little. Theories and research about motivation are expansive, so our purpose in this section is to focus on the most salient aspects so that you the teacher can consider how to best motivate your students.

Students will arrive in your classroom with different motivations. Some will have little interest or motivation to attend class, while others will have a high interest in learning your subject. On more than one occasion we have taught high school students who received money from their parents for the grades earned at school (e.g., $50 for an A, $40 for a B, etc.). At the same time, many students will be motivated to simply complete an assignment because they want it to be their best work. As we begin thinking about motivation, it would be interesting for you to consider what motivates you to do various tasks. These are often called *motivating factors.* What factors in your life motivate you? A few might be personal satisfaction, a sense of accomplishment, financial rewards, increased status among peers, or pleasing an instructor/supervisor. These types of motivating factors lead us to consider two different forms of motivation: extrinsic motivation and intrinsic motivation.

Extrinsic motivation is motivation brought on by external factors we believe will provide desirable outcomes, such as praise, a reward, or even avoidance

of negative consequences (Pintrich & Schunk, 2002). Extrinsic motivation is a means to an end; we are motivated to do something because the end result will be better. Teachers extrinsically motivate their students in a variety of ways: points for assignments, letter grades, praise for good behavior, assigning after-school detention for misbehavior, allowing free time if students all read silently for 30 minutes, or allowing students to revise their assignment for a higher score.

Intrinsic motivation refers to motivation to engage in an activity because of the satisfaction derived from the activity itself. Students who are intrinsically motivated view the learning process as valuable and genuinely want to understand the content. Teachers are able to encourage intrinsic motivation, though it is more challenging. Researchers suggest that the following types of activities tend to intrinsically motivate learners: challenging tasks that question existing understanding and create a sense of curiosity; tasks where the learner has some control over their own understanding; and tasks that evoke some aspect of our emotions (Lepper & Hoddell, 1989; Ryan & Deci, 2000).

Extrinsic and intrinsic motivation can both be used effectively in the classroom. Although it is important to realize that solely motivating one's students with extrinsic factors may promote learning for the sake of a reward (or for the sake of avoiding punishment), extrinsic motivation is not necessarily bad. For example, most people gain some motivation to go to work for the extrinsic reward of a paycheck. However, teachers need to consider how they might also provide opportunities to promote intrinsic motivation, where students want to learn because they genuinely enjoy the process. Researchers in this area are quick to remind us that motivation is neither intrinsic nor extrinsic exclusively (Pintrich & Schunk, 2002). For example, a student might study so she can receive a good grade and be eligible to play on the basketball team (extrinsic) and at the same time study because the course content is genuinely interesting to her (intrinsic).

While it is important to think about intrinsic and extrinsic factors, as a teacher it is equally important to consider what Brophy calls "motivation to learn" (1998). Brophy describes this as the "students' tendency to find academic activities meaningful and worthwhile and to try and get the intended learning benefits from them" (p. 162). Students who are motivated to learn may not have intrinsic motivation for all tasks, but they see the purpose and value of school. As a result, they are motivated to attend class, think about content, complete assignments, and engage in the activities because they believe that learning will be worthwhile. When you help your students understand how the content and activities are useful and beneficial you promote their motivation to learn. This might be accomplished by modeling to your students your interest in the content, and sharing how you motivate yourself to learn. Creating a classroom environment that exudes your enthusiasm for learning your subject, as well as your interest, care, and concern for your students' learning will both help develop your students' motivation to learn.

"Flow" Experiences and Motivation

An interesting approach for thinking about motivating students comes from a theory of intrinsic motivation called "flow" (Csikszentmihalyi, 1990, 1997). When someone is highly motivated from within, he is in a flow experience. This stems from a general belief that our abilities match the challenge of a task so well that we can simply enjoy the experience. During a flow experience we are unaware of time because we are so involved and concentrated on a task that we are only aware of the present. Additionally, we have confidence in knowing what needs to be done and how to accomplish it. This engagement creates a sense of calm and reward of its own. Most of the people who theorize about flow suggest that people can learn how to create a flow experience. Figure 1.1 below operates on two axes: one is the perceived challenge of a task; the other is a person's perception of his skill or ability to accomplish the task.

Consider how this figure might help you understand motivation and, more important, help you provide appropriate motivating factors for your students. When we perceive that a task presents a low challenge, we will respond with either apathy or boredom, depending on our perception of our skill or ability. If we see ourselves as highly capable, an unchallenging task will bore us. If we see ourselves as having low ability, it will lead us to not care. In contrast, a task that is perceived as highly challenging will promote anxiety (if we do not perceive our abilities or skills as being capable of completing the task), or it will promote a flow experience if we perceive a match between our ability and the challenge (Csikszentmihalyi, 1990, 1997).

This can be directly applied to the middle school or high school classroom in several ways. Low challenging tasks rarely serve to motivate students. According to figure 1.1 below, students will feel varying degrees of apathy and boredom; therefore, one needs to consider how to challenge them. If the task is so challenging that the students believe it is beyond their ability level, then they may be so distressed that they cannot complete the task. Students

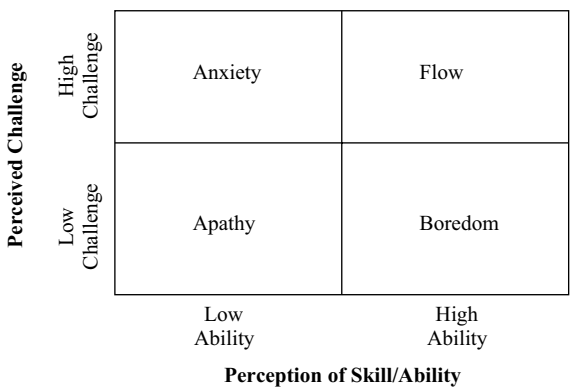

Figure 1.1

also need help in their perceptions. Tasks may appear too difficult when focus is only on the final product. However, when the task is broken into smaller, sequential activities, it becomes more manageable. Likewise, students may at first perceive themselves as lacking sufficient skill for a task. Often with proper modeling or "scaffolding" the students learn that they do have the skills after all. Consider the following examples, and possible ideas for moving students closer to a flow experience.

Boredom. A student who is an advanced reader may show little interest in reading an assigned book that is far below her reading level. She appears bored because the challenge is far below her ability. If the reading task is made more challenging, boredom will be less of a problem.

Anxiety. A shy student may decide to skip class on the day he is scheduled to make a 10-minute presentation to the class, because the stress of standing in front and talking is too great. If the student is coached on how to present, given opportunities to practice, or even allowed to present privately on videotape, he may perceive his ability to be more aligned with the task.

Apathy. A student in an art class may simply give up if he cannot accomplish the first drawing assignment of the year. He thinks that he will never be able to do well in the class, because he has such difficulty with the easiest task. If the teacher expresses her own challenges with drawing, and some of the strategies she has come up with, she provides a model of how to work through problems. If the student is provided with opportunities to practice on specific areas and master those areas he may see his skills improving.

Flow. A low-achieving student who loves to work on the computer is able to provide technical support to her classmates as they develop their personal websites. Classmates' challenges become her opportunities to see how well she can solve problems, and it makes her happy when others begin to rely on her to help with the assignment.

You will not be able to motivate all of your students for every task, but if you think of their motivation in these terms it will help you focus your effort to help them.

Self-Efficacy and Locus of Control

Self-efficacy is our expectation or belief about what we can accomplish as a result of our efforts. This in turn influences our willingness to attempt tasks, and what our expectations are (Dacey, Kenny, & Margolis, 2004, p. 173). Bandura has researched efficacy extensively, and his ideas can be directly applied to the expectations you might have for your future students:

> Self-efficacy beliefs determine how people feel, think, motivate themselves and behave. … A strong sense of efficacy enhances human accomplishment and personal well-being in many ways. People with high assurance in their capabilities approach difficult tasks as challenges to be mastered rather than as threats to be avoided. In contrast, people who doubt their capabilities shy away from difficult tasks which they view as personal threats. They have low aspirations and weak commitment to the goals they choose to pursue. When faced with difficult tasks, they dwell on their personal deficiencies, on the obstacles they will encounter, and all kinds of adverse outcomes rather than concentrate on how to perform successfully. (Bandura, 1994, p. 71)

Self-efficacy is developed through a wide range of factors, including family, peers, cognitive ability, and previous experiences. As a teacher you can work to increase a sense of self-efficacy by modeling successful strategies for students, providing opportunities for success, and helping your students develop strategies for overcoming obstacles. Similar to self-efficacy is the idea of *locus of control*. This is a concept that examines how much control people believe they have over events in their life. If you attribute success or failure to forces beyond your control (for example, you believe you were "just lucky," or "not sure why you failed") you express an *external* locus (location) of control. If you attribute your behavior to forces you can control (for example, you "studied hard for the test," or "won because of practicing") you have an *internal* locus of control. As a teacher, opportunities to help your students develop internal attributions will be beneficial, since control over their achievements/behavior will be from themselves and not from unpredictable outside forces.

Making It Work in Your Content Area

Partner with someone who will teach in the same content area as you. Generate examples of how students might behave in your classroom for each of the four student behaviors proposed on Csikszentmihalyi's flow (figure 1.1). For example, how might a student be apathetic in a geometry class, and how would he then behave? How might a student be anxious, bored, or in flow? For the three behaviors other than flow (apathy, anxiety, and boredom), come up with at least one approach through which you could modify either the student's perception of his skill, or the perception of the difficulty level of the task.

DIVERSE CHARACTERISTICS AND NEEDS OF MIDDLE SCHOOL AND HIGH SCHOOL STUDENTS

Your future classrooms will have students who are very diverse. When considering some of the ways in which students might be different from one another, the following provides an adequate but incomplete list: students will have different cultures, ethnicities, primary languages, economic backgrounds, religions,

family structures, physical abilities, and learning abilities. These differences are not exclusive, either. For example, some students who are English-language learners are also gifted, and some will have physical disabilities. In this section we will examine several aspects of diversity in your future students so that we can establish several key ideas. The diversity of your students needs to be considered (and embraced) whenever you plan lessons and activities. As a result, we will revisit this topic very specifically when the instructional strategies are examined in section 2 of this book. Your students will learn best when you are informed about their interests, ability and readiness levels, learning preferences, and other areas of their lives that allow you to maximize their specific learning potential (Tomlinson, 2004). As with any examination of adolescents, a caveat needs to be in place about exploring student diversity: generalizations about any group are not intended to create stereotypes. Since there are not universally applicable generalizations about teaching adolescents, it will be up to you to get to know your students as individuals. Diversity is a characteristic of your future students that should be enjoyed and considered for the range of ideas, viewpoints, and experiences that it will provide.

Ethnic and Cultural Diversity

A common term closely tied to diversity is *multicultural education*, which Banks and Banks define as follows:

> Multicultural education is a field of study and an emerging discipline whose major aim is to create equal educational opportunities for students from diverse racial, ethnic, social-class, and cultural groups. One of its important goals is to help all students to acquire the knowledge, attitudes, and skills needed to function effectively in a pluralistic democratic society and to interact, negotiate, and communicate with peoples from diverse groups in order to create a civic and moral community that works for the common good … *consequently, we may define multicultural education as a field of study designed to increase educational equity for all students that incorporates, for this purpose, content, concepts, principles, theories, and paradigms from history, the social and behavioral sciences, and particularly from ethnic studies and women studies.* (Banks & Banks, 1995, pp. xi–xii; emphasis added)

One implication of this definition is that multicultural education moves students' stories and experiences into the school curriculum. Instead of feeling as though they are not a part of the school and curriculum (often referred to as being "marginalized"), students from ethnic minority groups see their culture, history, and perspectives incorporated into the standard curriculum. In addition, students in the "dominant culture," or the culture of the social or political group

that holds the most power in a school and community, need to learn about and embrace stories, cultures, and experiences that are not the same as their own. Ladson-Billings has called teaching with students' culture and background in mind "culturally relevant teaching" (1995). An excellent place to read a summary of culturally relevant teaching is at the website of the Education Alliance at Brown University.[2] For example, the Education Alliance suggests seven principles for meeting the needs of ethnically and culturally diverse learners. Our summary of these principles are represented in the following overview.

PRINCIPLES FOR MEETING THE NEEDS OF ETHNICALLY AND CULTURALLY DIVERSE LEARNERS

1. **Positive Perspectives on Parents and Families:** Parent involvement should not be limited to participation in school functions. Often, religious and cultural differences preclude active participation in school activities. However, parental involvement also includes how parents communicate high expectations, pride, and interest in their child's academic life (Nieto, 1996). Teachers should try to understand the parents of their students, because that will often provide insight into the student at school. This requires communication with parents through such avenues as phone calls and e-mail, school visits from the parent, home visits from the teacher, and even informal conversations at community events. Additionally, teachers can help inform parents of the various social and academic services offered by the school and by organizations associated with the school. If you have not had experience with the culture of a parent, you should learn about cultural activities and beliefs. This will provide a beginning level of knowledge that will help you begin to communicate with parents.
2. **Communication of High Expectations:** All of your students should hear that you expect them to attain high standards in their schoolwork. This message must be delivered by teachers, counselors, coaches, administrators, and other school personnel. Teachers should understand students' behavior in light of the norms of the communities in which they live, which is why the first point above describes the importance of talking with parents.
3. **Learning Within the Context of Culture***:* Schools have a dominant language. Typically this is the standard American English that is common in middle and upper socioeconomic groups. Your students who do not live in a home and/or culture that is close to that of the schools may be inhibited from learning because of "language barriers." Other barriers also exist, including preferences for learning. Some cultures emphasize speaking while others value silence. Some cultural groups put value on cooperation and others do not, preferring individual work. By gaining knowledge of the language and culture of your students you will be able to create lessons that promote learning because they will use contexts with which your students are familiar.
4. **Student-Centered Instruction:** During student-centered instruction, learning is a cooperative venture; students are more engaged in learning the content than when the teacher is the center of attention. This may entail everything from brainstorming topics, to student-led discussion groups, to cooperative learning projects, to

service learning activities in the local community. These types of activities provide opportunities for students to develop their own ideas, and receive feedback on these ideas. Students from diverse backgrounds are able to bring their ideas and cultural perspectives to the classroom, and to work on projects that are culturally and socially relevant to them.

5. **Culturally Mediated Instruction***:* Culturally mediated instruction and learning occur in an environment that encourages multiple viewpoints and allows for inclusion of knowledge that is relevant to the students and to their diverse cultures and backgrounds. Your goal should be to create an environment that encourages and embraces culture by using management and learning strategies familiar to students' culture, provides students with opportunities to share their cultural knowledge, and even questions and challenges students' beliefs and actions. Students should learn that there are often multiple ways to interpret a statement, event, or action. By being allowed to learn in different ways or to share viewpoints and perspectives in a given situation based on their own cultural and social experiences, students become active participants in their learning (Nieto, 1996).
6. **Reshaping the Curriculum***:* The curriculum should use resources other than textbooks for study. For example, students could research a topic within their community, interview members of their community, or provide information to other students on alternative viewpoints or beliefs on a topic. Students' strengths in one subject area will support new learning in another. If your students use their personal experiences to develop new skills and knowledge, a more meaningful connection between school and real-life situations is established (Padron, Waxman, & Rivera, 2002).
7. **The Teacher as Facilitator:** As teachers, you should develop a learning environment that is relevant to and reflective of your students' social, cultural, and linguistic experiences. You will act as guides, mediators, consultants, instructors, and advocates for your students, helping to effectively connect their culturally based and community-based knowledge to the classroom learning experiences. Teachers should use the students' home cultural experiences as a foundation upon which to develop knowledge and skills. Encourage students to share artifacts from home that reflect their culture, write about traditions shared by their families, or research different aspects of their culture. Teachers should also work to accommodate learning preferences of their students that are linked to cultural background and take into account varying levels of English-language proficiency. This might also occur by using a community speaker, or inviting parents who can demonstrate alternative ways of approaching a subject or a problem to class (Ladson-Billings, 1995; Padron, Waxman, & Rivera, 2002).

Exceptionalities

In your future classrooms you will teach students with unique needs, talents, and interests. One challenge you will face will be to differentiate the curriculum so that every child has opportunities to learn in an environment that is safe and least restrictive. What this means for you is to consider how you might make your daily instruction individualized for each of your students. You can

manage this task for many of your students, but you will also have students who will need to receive special services—or special education. These students are frequently referred to as "exceptional students," and teachers have a legal, and we believe ethical, obligation to provide lessons to them that will promote maximum learning. Culatta and Tompkins (1999) developed this list of exceptionalities your future students may have: mental retardation; learning disabilities; behavioral or emotional disorders; communication disorders; physical and/or health impairments; hearing and or vision impairments; severe or multiple disabilities; and giftedness.

Exceptional students may receive special services from an instructional assistant who attends classes with them, or from courses and programs that pull a student out of the general education classroom. When possible, however, exceptional students are kept in the general education classroom. This is known as the "inclusive model" because the general education classroom is a place that includes all students. It is easy to understand that many educators disagree about how to educate children with exceptionalities. Proponents of a fully inclusive model speak to the benefits that all students have when in a room with diverse learners. Students of all capabilities work together to help one another learn. The challenge for teachers is to provide opportunities for learning that account for the wide range of students. Opponents of inclusion suggest that placing students in programs separate from the general education classroom allows for more individualized instruction and learning; they argue that the burden on the classroom teacher will be too great if all students are placed together. The challenge for these "pullout" programs is to ensure that academic expectations remain high. Additionally, boys are identified as needing special services more than girls, as are students of color and students from homes that are disadvantaged economically. Whether these students should be taught separately, included in the general classroom, or taught with some combination of these two is a raging debate in the schools. In any case, you will need to consider how you will help students who have exceptionalities learn when they are in your classroom.

One of the most powerful tools that you will have as a teacher in an inclusive classroom will be your exceptional students' individualized educational program, or IEP. When a public school student needs special services, an IEP must be designed. These are developed by the special education teachers/specialists in your school, with the help of parents, other teachers, school personnel, and at times, the student. An IEP give details about the educational supports and services that will help the child with a disability receive appropriate instruction. An IEP may specify instructional strategies, strategies for managing behavior, motivational techniques, and other insights about how to best help the student learn. Test scores, physician reports, and psychological, cognitive, and/or other medical reports might also be included. By contacting your special education expert(s) in the school you will have powerful strategies for meeting the needs of your exceptional students. New regulations emphasize that the IEP team must consider a student's strengths as well as

areas of weakness when formulating an educational plan. An IEP is required by federal law whenever a student has a recognized disability. In 1997 the Individuals with Disabilities Act was passed into law, specifying disabilities and strengthening the expectations of learning for all students. Each state, or in some cases, each district or each school, designs its own IEPs for its students with disabilities. Some students do not have disabilities that require special education services. They might not receive an IEP, but they could receive a 504 Plan, which is a program of instructional services for assisting special-needs students who are in a regular education setting. A student who has a physical or emotional disability, who has an impairment that restricts one or more major life activities (e.g., caring for oneself, performing manual tasks, walking, seeing, hearing, speaking, breathing, working, and learning), or who is recovering from a chemical dependency should receive a 504 Plan.

The instructional strategies that we will examine in section 2 of this book can be adapted to meet the individual needs of students. They allow for all students to learn, whether they are considered general education students, students with learning disabilities, or gifted students. Specifics for differentiating the instructional strategies in order to help your students learn most effectively are described for each strategy.

TEACHING ENGLISH TO STUDENTS OF OTHER LANGUAGES

You might relate to the following reflection from a recent student teacher:

> My university supervisor came to my World History class one day to observe. After class he was encouraging me to consider ways I could include all the students and not only the ones that volunteered to speak-up. He asked if I noticed the three students sitting in the back corner that didn't say a word the entire class. I did notice them. They speak Ukrainian and I had no idea what to do.

The demographics of the United States, including the number of those that speak languages other than English, are changing rapidly. Teaching students that are learning English as their second or third language is not new but it has become increasingly common. Consider this statistic: the total enrollment of U.S. students in grades K–12 grew by 9.5% between 1985 and 1995. At the same time the number of students in the United States with limited English proficiency (primarily immigrants) grew 109% (Olsen, 1997). This trend continues today. A much larger percentage of new students with limited English skills than new students with English as their primary language are enrolling in schools. Statistics that reflect school population change have been used to support what has been referred to as the *demographic imperative*. It is imperative for us to prepare ourselves to teach all of our students.

Obviously, not every new teacher will choose to become a certified expert in teaching English as a second language; however, there are several ways to prepare. An important start is learning about the four basic elements for engaging students who are not proficient in English. First, it is important to be able to understand some basic terms used so that when you are placed in your internship or begin your first teaching position you can ask informed questions. The terminology can be tricky, if not head spinning at times. A person could walk into the teacher's lounge and hear talk of ESLs, TESOLs, ELLs, ESOLs, bilinguals, emergent bilinguals, and LEPs. The following glossary of terms will prepare you with some key words and phrases.

A BRIEF GLOSSARY FOR TEACHING STUDENTS WITH ENGLISH AS A SECOND LANGUAGE

BICS: Basic Interpersonal Communicative Skills. This might be thought of as conversational language. A distinction between BICS and CALP was introduced by Cummings (1979). See CALP.

Bilingual: An ability to speak two languages.

Bilingual education: An approach to instruction that includes the continued development of the student's native language, the learning of a second language, and the use of both languages to teach academic content.

Biliteracy: An ability to read and write in two languages. See Freire (1978) for information about the implications of biliteracy or literacy on issues of social justice.

CALP: Cognitive Academic Language Proficiency. Immigrant children typically take 2 years to acquire a conversational fluency (BICS) in a new language. It may take the same child at least 5 years to reach an academic proficiency (CALP) in the new language.

ELL: English Language Learners or students who are not proficient in English.

ESL: English as a Second Language. A program or system of instruction designed to teach English to those that speak other languages. This system typically will use academic content and is usually thought of as a component of bilingual education.

ESOL: English Speakers of Other Languages. Some prefer this to ESL because it recognizes that English could be the third or fourth language of the student and that this should be acknowledged. For example, a student from Mexico may speak Mixteco (a Native language) as his first language and Spanish as his second language.

Immersion: Technically immersion is a bilingual program developed in Canada where 90% of the early elementary school student's day is in the minority language (the non-mainstream language). Many times, however, this term is used in school to refer to placing a student in a mainstream classroom (not necessarily as part of a well-thought-out program).

LEP: Limited English Proficient. This is usually used as an adjective to describe students, as in "the LEP student." Some prefer not to use this term for the same reason some prefer not to use the word "disability" — it focuses on what a student cannot do instead of what he can do.

TESOL: Teaching English to speakers of other languages OR the professional organization Teachers of English to Speakers of Other Languages.

Second, it is important to know that you are not alone. There usually will be a teacher in your school that specializes in teaching ESL. Her program could look like one of the many possible varieties of teaching English but most likely will reflect current trends of teaching ESL. This means that students of all levels of English proficiency come to her class to either learn the academic subject (ESL content class) or have class time to support them with their assignments from mainstream monolingual English classes; support may be in English or in their first language, depending upon obvious variables. Of course, some students will attend introductory-level English classes as well. This is important to keep in mind, but—without question—conscientious teachers will not shift all of the responsibility for the learning of an ELL onto the ESL teacher.

A third element to consider is the way people acquire new languages. Having a basic understanding of language acquisition theory will help as you try to design or modify lessons. While taking a course on second language acquisition theory would undoubtedly benefit any teacher, here we will focus on two of the components that influence language acquisition in a school context (Cummings, 1997; Thomas & Collier, 1997). Students learning English will tend to begin by learning social language. They learn what they use in everyday conversations. Immigrant children typically take two years to acquire a conversational fluency (BICS) in a new language. Students take much longer to negotiate their way into the more formal academic language. It may take the same child at least five years to reach an academic proficiency (CALP) in the new language. The catch is this: an unknowing teacher may assume that because he hears a student speak English during class that the student is equally proficient in academic language, the language of teaching and learning. This will most likely not be the case. In addition, because textbooks are written in academic language the student's actual proficiency could easily be overestimated and the student not be provided with adequate assistance for understanding what he or she reads for class. Another consideration for teachers is that some students may not have a literacy base at all. In other words, they may not read or write in their first language. This obviously would have an impact on their ability to read or write English even if they are verbally performing well.

The final element for you to consider is inclusive methods of teaching. In the following chapters we will help with ideas for taking what we know about students of ESL and moving into practical ways to include these students in the mainstream class.

Gender Considerations

Campbell and Steinbrueck (1996) write that teachers often communicate different expectations of male students than they do of female. Teachers often give male students more attention and provide them with more opportunities to share during class activities. Others have observed teachers and conclude that teachers tend to praise boys for the intellectual contributions to class, and praise girls for their proper behavior in class (Sadker & Sadker, 1982, 1994). As a result, teachers may reinforce societal norms about gender-biased behaviors that actually limit girls' academic performance and enhance boys' achievements. In recent years a series of programs have encouraged girls' participation in mathematics and science classes, and increases in girls' performances in these areas are starting to be realized. Concern is now beginning to rise about the low number of male students who attend colleges, with some estimates suggesting that in the near future two female students will graduate from college for every male student. Suffice to say that gender differences exist, and you need to reflect on your interactions with students to ensure that gender bias is not affecting your students' opportunities to learn.

The conclusions of Pipher (1994) are compelling and sobering: in early adolescence, girls' IQ scores drop and "their math and sciences scores plummet. They lose their resiliency and optimism and become less curious and inclined to take risks" (p. 19). This could partly be due to changes from the transition from elementary school to middle school, changes brought about by puberty, and/or increased awareness of peer and societal influences on their behavior. Boys are changing as well, repressing emotions and feelings and even using aggressive behavior in order to fit in with societal expectations (Sadker & Sadker, 1994; Thorne, 1994). Boys in middle school and high school engage in more "at risk" behaviors (gang membership, drug use, physical daring, etc.) than girls. Boys tend to receive lower grades, and make up the majority of students referred for special-education services. For example, 71% of students identified as learning disabled and 80% of those identified as emotionally disturbed were boys (Sadker & Sadker, 1994).

For these reasons and more it will be important for you to consider the role gender plays in your content area. Poetry is perceived by many high school students as being feminine even though famous poets throughout history have been men; mathematics is often perceived as being masculine even though women have made significant contributions to that discipline. Women's roles are often omitted from historical accounts; men's conquests dominate the conventional stories. Check your curriculum to identify if you are missing important opportunities for clarifying misrepresentations of women and men, or if you are underrepresenting the roles of women or men in the content your students are studying. When teaching, reflect on the nature of your interactions with students. If you teach in accord with the samples from many studies that McCormick examined in 1994, you will likely give boys five times more attention than girls. Work to provide gender equity in your classroom.

Poverty in Schools

According the United States Bureau of the Census, 37 million Americans are living in poverty. Of these, 13 million are under 18 years of age.[3] The socioeconomic status of your students will likely impact their learning and the educational resources available to you. The National Commission on Teaching and America's Future has issued a report that supports over twenty years of research on poverty and schooling. The commission concludes that

> children at risk, who come from families with poorer economic backgrounds, are not being given an opportunity to learn that is equal to that offered to children from the most privileged families. The obvious cause of this inequality lies in the finding that the most disadvantaged children attend schools that do not have basic facilities and conditions conducive to providing them with a quality education. (NCTAF, 2004, p. 7)

Gorski (2005) follows up on this report, noting that

> schools with large percentages of low-income students are more likely than schools with large percentages of wealthy students to have an abundance of teachers unlicensed in the subjects they teach, serious teacher turnover problems, teacher vacancies and large numbers of substitute teachers, limited access to computers and the Internet, inadequate facilities (such as science labs), dirty or inoperative student bathrooms, evidence of vermin such as cockroaches and rats, and insufficient classroom materials. (p. 3)

Many implications can be derived from these accounts. It is important to recognize the impact of poverty on schooling, both in terms of educational opportunities for your students and the support systems available to them outside of school. You will want to consider whether your assignments and activities unfairly allow students with more resources to have an advantage over those with fewer. Teachers used to be discouraged from finding out the socioeconomic status of their students lest it bias them. Today, the thinking is that teachers must be aware of student differences so they can help each child uniquely. When you know students are living in poverty, you then check if you are assuming that all students come to school from the same background. Poverty is systemic, and it is difficult for schools and teachers to devise a list of strategies for overcoming it. We have developed the following list of topics related to poverty and wealth that we hope you will think about when you are planning your classroom activities. This is not an inclusive list, but a starting point for considering economic diversity in your classroom.

- Some students work for pleasure, some to help pay bills, and some will not work at all.
- The value of education varies dramatically in the homes of your students.
- Opportunities to travel, have access to materials and resources, and use the Internet outside of school are correlated with higher economic status.
- Parent/guardian availability to help with students' schoolwork is widely varied.
- Some students have nutritious breakfasts and lunches, while others will eat little or eat poorly.
- Crime rates and other safety issues could impact students' opportunities to complete schoolwork.

Making It Work in Your Content Area

Think of three specific strategies that you might use in your future classroom that will raise the status of students who are typically marginalized due to ethnicity, exceptionality, or gender. Consider the following examples to get you started thinking about this: If you are a woman who plans to teach mathematics, then you could tell girls how you overcame any stereotypes about women and mathematics. If you plan to teach American literature, you might select a story by Alice Walker because African American women are often underrepresented. If you plan to run a mock congress in your civics course, you might consider having students who are not in leadership roles at your school hold offices of class president or ambassador during the simulation.

CONTEXTS OF TEACHING IN A CHANGING AND TECHNOLOGICAL SOCIETY

You will find that in educational circles technology is defined in many different ways. Some, such as the Association for Educational Communications and Technology (1994), define educational technology very broadly so that it includes procedures and resources for learning. This could involve the process one uses while including a tool for learning as itself a type of "technology." Others might define it almost exclusively as the use of resources as tools for learning. Tools might include many resources, from common pencils to sophisticated systems for interactive communication (e.g., laptop computers or cell phones). For the purposes of this text, we will be using the term *technology* to specifically refer to modern electronic resources found in classrooms that can be used for teaching and learning. Most often we will be referring to digitally oriented resources used by and found on computers.

The technology available has changed dramatically over the last twenty years. This is obvious to many of us, but to others that have grown up with

the changes it may go undernoticed. The frantic pace of technological change is an unremarkable way of life to many middle and high school students of today; these students have spent their entire lives learning new software, playing video games, updating their computer operating systems, and looking forward to the latest programs with state-of-the-art audio and graphics. Some have described these students as native speakers of the digital environment, or "digital natives." That, by extension, would make those not born into the digital world "digital immigrants"—those trying to grasp the myriad technological changes, those that tire of repeatedly learning new versions of software, those that speak with digital "accents," and those that, according to some, think differently (Owston, 1997; Prensky, 2001).

As long as we are considering this metaphor, it is important to note that some have purposefully decided to stay in the "old country" and not (or try not to) "immigrate" at all. These note how technology does not always have a positive impact on our society and, in fact, takes more than it gives (Oppenheimer, 1997; Postman, 1995). We believe that just as it is important to consider ways to include ESL students while planning lessons, it is also important to consider ways to include the digital language of our students, as well as to take into account our own fluency in that language. Some teachers are naturally apprehensive about using the new language (a computer), while others do not realize that they are speaking with a "heavy accent" and that their students are the "native" speakers (i.e., highly skilled with computers). In addition, it is important to remember that asking your students to use technology implies that students will need access to computers and other electronic devices. Much has been written about the "digital divide," the socioeconomic gap between communities and people who have access to computer technology and those who do not. For example, if you assign students to access the Internet as part of a homework assignment, you are assuming that all students can do this outside of school. If some can, and some cannot, then you are placing students who do not have access at a disadvantage. This is not equitable. Care needs to be taken when implementing student activities tied to technology.

Through the years, as we have discussed using technology with teachers and aspiring teachers, there has been common agreement: most teachers do not want to be "technologists." Teachers are less interested in learning about the gadgets than learning how they can be used in a lesson. One of the most important things a person can discover about using technology for teaching and learning is that the focus in the classroom should not be on the technology, but on how it can be used to promote student learning. "Digital immigrants" can be a skeptical group because most of us have seen the well-meaning use technology in a manner that adds little value to a lesson or, worse, becomes a distraction. Adding technology has become the easy part of teaching, but not necessarily the best part. The question we ask ourselves should not be "How can I infuse technology into this lesson?" but instead "Can value be added to the lesson by using technology, and if it can, then how?"

We have also seen technology used in well-thought-out, valuable ways. Some have argued that technology helps increase student motivation. Of course, any teacher would be interested in something that is motivating. Unfortunately, motivation due to the use of a new gadget is somewhat shallow and often expensive. However, motivation due to increased control of one's own learning is empowering, and technology can help facilitate this. But this takes a rethinking of the way we teach, not simply the addition of more technology to our classes. Technology can also be used to promote, establish, and support dynamic instructional strategies that are designed to meet learning objectives. In fact, in a review of cases studies, the CEO Forum (2001) concluded that technology has the greatest impact when it is integrated into the curriculum for the purpose of achieving clear, measurable educational objectives. In the chapters describing instructional strategies that follow in this book, we show how technology could play an important role when using specific instructional strategies to help students learn.

Making It Work in Your Content Area

Think critically about the use of technology in the classroom. Go to the technology-infused lesson-plan database created by in-service teachers in Mississippi at CREATE (http://www.create.cett.msstate.edu/create/classroom/lessons.asp). Choose a lesson to review from the areas of language arts, science, math, and social studies.

What technology is to be used? Do you think the technology will help students learn the content more effectively? If so, why? If not, why did the teacher use technology and what could the teacher have done differently? What caveats would you give any teacher that is interested in trying the lesson?

GUIDING PRINCIPLES FOR CLASSROOM MANAGEMENT AND DISCIPLINE

After many interviews with teachers (both preservice and in-service), we have concluded that classroom management and student discipline are closely linked to the instructional strategy one is using. Each instructional strategy has its own management concerns/issues; the chapters herein on instructional strategies specify how to best manage the classroom while using a particular strategy, and how you might effectively discipline students during each strategy. Therefore, the chapters in section 2 build on the management and discipline principles that we will examine in this section of chapter 1. We intend for this section to provide some general terminology and overviews of key components for managing an effective learning environment.

One of the greatest worries in the minds of many student teachers has to do with management and discipline problems. Preservice teachers often wonder if they will be able to maintain order, earn student respect, oversee group activities, or effectively handle all of the logistical and organizational tasks of the

classroom. As well, specific questions arise from working with teachers, such as "Am I really not supposed to smile until December?" or "What should I do if a student says 'no' when I ask him to do something?" These are very real concerns. Many worries fade when you meet your students, but new concerns often arise. Effective teachers are able to manage the classroom and discipline students so that optimal learning can occur. In addition, teaching will be much more enjoyable for you when the classroom is a community of learners in which you and your students are comfortable.

Differences Between Management and Discipline

When preparing to help students learn, you will use a series of instructional strategies. While these strategies will help you determine appropriate ways to assess your students' learning, they will also provide guidance as you identify the activities students will engage in, select materials and resources that you and the students need, and consider the logistics of having the day's events work together to promote student learning. The best-planned lessons can be unsuccessful, however, if you do not think through how you will manage the learning environment, and if you are unable to effectively discipline students who are behaving inappropriately.

We differentiate between classroom management and discipline as follows: *classroom management* is a proactive approach to helping students learn, and *discipline* is a reaction to student misbehavior. Classroom management addresses all of the aspects of the classroom that will promote student learning. You are overseeing all that happens in your classroom. Many educators refer to the classroom as a "community of learners" in which you and your students are committed to making the classroom environment a safe place where knowledge, skills, and dispositions are developed over the school year. As the adult leading that community you need to ensure that students are supported, cared for, challenged, and meeting the learning outcomes for your course. Classroom management involves overseeing the classroom so this happens. *Discipline* specifically refers to how you can effectively control students so their behaviors maintain the sense of community in your classroom. Part of discipline enables students to self-monitor their behaviors (so they literally become self-disciplined), but part of discipline involves the specific action that teachers take when a student disobeys a rule or fails to meet an expectation. While teachers need to discipline students, we suggest that many problems arise simply because the classroom is poorly managed.

Consider the following very real dilemma that a teacher might face: Students are seated in groups of four around the room. One group of students needs the teacher to help in understanding the assignment. A second group of students, on the opposite side of the classroom, is being disruptive and not working on the assignment. The teacher knows that her proximity has a significant impact on disruptive behavior; that is, when she is in close proximity

to the disruptive group they quiet down and work. However, this keeps her away from the group of students needing her help. What could the teacher do to manage this situation? How could she address this proactively? What discipline issues are at work?

Relational Teaching

Many conflicts and problems can be avoided if teachers try to build appropriate relationships with their students. The term we use for this is *relational teaching*. As the term implies, relational teaching places high value on the relationship the teacher has with the whole class as well as with each student. Rather than thinking of your students as a group of adolescents that come to your classroom, you think of them individually, with each possessing their own unique experiences, struggles, strengths, and weaknesses. During the school year you look for opportunities to build teacher/student relationships with each member of your class. This takes time and effort, but it is surprising how quickly you can get to know your students. Benefits from a relational teaching approach include: knowing how to assign students to groups because you know each students' strengths and weaknesses; understanding pressures and influences on a student that may affect his or her behavior in class; developing a sense of mutual respect for one another; reducing "acting out" or misbehavior by students because they do not want to harm the relationship you have developed; and, increasing interest in the development and learning of your students as you become more invested in their lives.

Relational teaching also has to do with developing what is often called "with-it-ness." When teachers are "with it" they are described as being keenly aware of all that is happening in the classroom. Sometimes it seems as though a "with it" teacher has eyes in the back of his head, or has an extraordinary sense of hearing and can hear comments whispered on the far side of the room, or is able to preempt a potential problem before it starts. A significant part of being so tuned in to your students that you are "with it" requires developing a positive rapport with them.

Doda, George, and McEwin (1987) devised 10 "truths" about effective schools. These truths attempt to identify how teachers can help to provide a school environment that is effective and helpful to students, and these truths reveal the importance of teacher-student relationships.

Ten Current Truths About Effective Schools

1. Work to reduce the size of student school groups so that as a teacher you are able to build relationships with students.

2. Remember/emulate the climate and tone of the elementary school. Students' work should be on display, and lots of visual stimuli present to promote thinking about behaviors and contents.
3. Spend time with students in noninstructional ways. For example, spend time with them at sports events, as club advisors, or when you see them in the community.
4. Ample rewards and awards help the students feel they are being appreciated for their thinking and work.
5. Reduce the isolation and independence inherent in teaching by interacting with other teachers, parents, and students. See teaching as a collaborative effort among many constituent groups.
6. Do not sit down while you teach. Rather, monitor the classroom by interacting with students and assisting them one-on-one.
7. Make lessons "real" and "hands-on" so students have opportunities to engage in genuine learning experiences.
8. Use/develop your sense of humor. Avoid sarcasm, but be willing to laugh at tense situations and even at yourself.
9. Establish big/high/rigorous expectations for students, but plan and teach in small segments.
10. Look for opportunities to develop a strong rapport with each student.

Approaches to Classroom Management

Every teacher develops his own approach for managing a classroom. A successful teacher will often claim that he does not have management or discipline problems, but when you watch him teach he is continually monitoring and interacting with students to ensure that they are acting prosocially and adhering to classroom expectations. If a student begins to fall asleep, the successful teacher casually walks by and taps her on the shoulder. If two students are passing notes, the teacher does not draw attention to the behavior but simply continues talking while standing next to the two students. His proximity makes them put the notes away. If a challenging assignment is beginning on Monday, the successful teacher has anticipated student questions and is able to address most of them when the assignment is described. In this way students feel as though their ideas and worries are considered. In each of these situations the teacher manages the learning environment with an approach that works. In this section we will describe seven effective and long-standing approaches to classroom management. What you end up using will undoubtedly be a hybrid of these, but an initial examination of these seven will allow you to start thinking about effective management approaches.

Behaviorism Skinner did not describe an approach to classroom management, but the principles of behaviorism that he established provide some

general ideas. The most useful principle for management is *behavior modification*. Behavior modification operates on positive and negative behavior. True to the stimulus-response of behaviorism, a teacher notices a student behaving properly and acknowledges, praises, or in some way reinforces that behavior. The reinforced student, along with his classmates who witnessed the reinforcement, will likely repeat this behavior and perceive it as beneficial. Students who are misbehaving are not acknowledged, and in an effort to earn praise also behave appropriately. Students receive external rewards for behaving appropriately; these often take the form of tokens or other tangible items. For example, a teacher might put a pencil in a mug whenever all of the students are sitting down before the bell rings to start the period. When the mug is full (about 15 pencils), the class receives a "free day" to learn about a topic of their choice, or even to have a pizza party. Rules are established and rewards are provided for following them.

Canter's Assertive Discipline Model This model (Canter, 1976; Canter & Canter, 1992) provides a set of clear-cut expectations for the classroom. Students can expect teachers to provide a safe learning environment, and teachers can expect students to pay attention and not disrupt the class. The teacher establishes clear expectations, justifies why these expectations are needed, and lists the consequences for misbehavior (not meeting expectations). Once established, the teacher expects adherence to the classroom rules. Students have the choice to behave in accord with the rules or to misbehave. Misbehavior is considered to be a poor choice, and the teacher must be assertive and follow through with the stated consequences. This will teach students to behave responsibly in the classroom. When helping students remember the expected behavior, the teacher should state the expectations to the student ("Chris, remember to bring your book to class"), or to praise a student in front of the class for behaving properly ("Chris remembered to raise his hand. Nice job!"). If a student perpetually misbehaves, then the teacher "moves in" close to the student, establishes eye contact, restates the expectations, and indicates the next consequences of continuing to misbehave.

The Dreikurs/Albert Model Dreikurs (with Cassel, 1972) suggests that the classroom needs to be democratic, but his ideas became more applicable after Albert (1974) refocused them specifically to classroom management. This has implications for how the rules are decided and for the instructional strategies that are employed. Students impose limitations on themselves for the purpose of directing behavior so that it promotes learning. The teacher and students cooperate to determine classroom procedures, rules, and even the discipline for misbehavior. Democratic teaching is very different from "autocratic teaching" (where the teacher is always in charge of the classroom policies and disciplining students, and where students have no need to develop self-discipline), and it is different from "permissive teaching" (where the teacher does not help enforce the classroom policies and students do not learn about responsibility,

accountability, and self-discipline). The democratic teacher engages students in the process, and fosters a sense of self-discipline among the students. Students have goals that they strive to attain, one of which is a sense of belonging to the classroom community. If a student does not follow the class-developed rules, then authentic consequences result. Dreikurs suggests that misbehavior is the result of a student seeking after one of four wrong goals: (1) getting attention; (2) seeking power; (3) getting revenge; or (4) withdrawing and not participating. As the teacher manages the classroom she looks for students who are pursuing these wrong goals and then redirects them by enforcing the natural consequences of breaking rules.

Ginott's Model This model (Ginott, 1972) is built around the key term of *congruent communication*, or communication that is about situations rather than student character or personality. In other words, instead of telling students that they are being rude by interrupting during a classroom discussion, the teacher might say,"This is a whole-class discussion. We need to allow everyone a chance to express their ideas." Students' behavior, thus, is managed by addressing the situations that students are in and inviting students to cooperate. In the teacher's comment, the first sentence addresses the situation (a class discussion), and the second sentence asks for cooperation (letting everyone speak). When the teacher does not attack a student's character or personality, the student may be more willing to discuss his feelings about a situation in the classroom. This open line of communication, along with the teacher modeling appropriate behavior, enables the teacher to change the attitudes and beliefs of students and not merely their behavior. This takes time, and depends on repeated opportunities to communicate with students. Ginott feels strongly about the powerful role that teachers play in students' lives, writing,

> I am the decisive element in the classroom. It is my personal approach that creates the climate… . As a teacher I possess tremendous power to make a child's life miserable or joyous. I can be a tool of torture or an instrument of inspiration … in all situations it is my response that decides whether a crisis will be escalated or de-escalated, and a child humanized or de-humanized. (p. 13)

As a result, Ginott suggests that communication without attacking the students' character will lead to a well-managed classroom.

Glasser's Model According to this model (Glasser, 1992), the teacher expects all students to produce quality work. Teachers must lead students in learning activities that students and teachers agree are worthwhile and useful. This model suggests that students want basic needs met: survival, belonging, power, fun, and freedom. Survival is not always tied to the school (though it can be if the student feels unsafe), but the other four needs can be met at school. When these basic needs are not met, students are frustrated. When they are met,

students are satisfied. If students' needs are not being met, the frustration leads to apathy, and the learning is seen as irrelevant. Teachers must manage the classroom to create an environment that has some fun, allows students a certain amount of power, has some degree of freedom, and helps students feel part of the classroom learning community. This can be partly accomplished when the teacher has high expectations for quality work, and provides opportunities to learn course content in depth. Students then need encouragement to determine their own paths for attaining this level of quality. The teachers role is to encourage quality work by challenging students to problem solve, building relationships with students, and continually being available to help. This can be done individually or during class meetings where expectations for quality work are further refined.

Jones's Model This model (Jones, 1987) differs from the others in its emphasis on nonverbal communication. The teacher's "body language, facial expressions, gestures, eye contact, and physical proximity" (Charles, 1996, p. 128) all send important messages to the students and help manage the learning environment. The messages from teachers to students about their behavior can be very powerfully sent without saying anything. Some teachers allude to this when they proclaim that when they simply give their students "the look" everyone knows it is time to behave. The importance of proximity is also highlighted by Jones. By simply standing near students who are misbehaving in some fashion, a teacher is able to quickly end the behavior without drawing a lot of attention to the situation. This model also suggests that teachers provide incentives for students that are genuine and will motivate all students in the class. Incentives such as extra credit will motivate select students, while an ice cream party will likely motivate all students. Tied to this is Jones's suggestion to follow "Grandma's Rule": First eat your vegetables, and then you can have your dessert. Translated to the classroom, teachers should remember that students should behave properly before receiving any rewards for their behavior. Teachers must not reward students based on students' promises to behave well.

Aside from nonverbal communication, Jones suggests that teachers should not spend too much time with any one student. Part of managing the classroom is to move around to all students. If a teacher spends too much time/attention on one student, the other students in the class are not monitored—for behavior or for understanding. Also, students need opportunities to work through academic problems without building dependency on the teacher to answer every question. Jones suggests that teachers should provide efficient help to students, and limit interaction to 20-second intervals. He suggests three steps: (1) acknowledge what the student has done correctly, (2) suggest what needs to be done next; and (3) leave the student to work on the details.

Kounin's Model Kounin's (1977) ideas about classroom management center on teacher behaviors that engage students in the lessons. He suggests that

misbehavior rarely occurrs when students are actively learning. Kounin's principal concepts are succinctly described by C. M. Charles (1996, pp. 45–46):

The ripple effect. This is a phenomenon in which the teacher's words or actions are directed at one student, but spread out and affect behavior of other students. Similarly, if misbehavior is ignored by the teacher, the ripple effect could spread a belief that the teacher will not enforce certain rules.

With-it-ness. As described earlier, this term to refers to a teacher's knowing what is going on in the classroom at all times. Students do not think they can "get away" with anything if their teacher is "with it."

Momentum and smoothness. Lessons need to keep moving by not spending too much time on details ("overdwelling"), having smooth transitions from one lesson to another, and varying class activities to keep students engaged.

Student accountability. This term refers to the teacher's efforts to keep students alert and involved in the lessons. This may involve calling on students to answer questions, or other requests that require students to be attentive.

Overlapping. This is Kounin's term for multitasking—the ability to attend to two or three tasks at the same time. For example, the teacher is able to write on the board while answering a question and redirecting a group of students working on a group project.

Satiation. This means that students are done thinking about a topic, and that the teacher needs to change his approach. Satiation can be brought about by boredom, repetition, or frustration.

Valence and challenge arousal. This is Kounin's term for how teachers might make activities more enjoyable/challenging. Variety of activities, teacher enthusiasm, and multimedia presentation are ways to keep students engaged (and not satiated).

Managing the Learning Environment

As we mentioned earlier, exuding a sense of being "with it" and using your proximity to students for making your presence known will help a classroom environment. Establishing classroom rules and routines will also help you manage the learning environment. Rules might be established by students, the teacher, or in concert together, but they need to be consistently adhered to by everyone in class. It is more effective to state your expectation in terms of what should be done, rather than what you do not want. For example, the rule "Do not be late to class" is not as effective as "Be ready to start class when the bell rings." Or the rule "Do not interrupt" is more clearly stated as "Let a classmate finish her thought before you talk." In work that we have done with teachers, it appears that fewer, more general rules and expectations are more effective than

many specific rules. "Respect other's opinions" is vague, but it may be a better rule then 10 specific rules about respect. If you are going to ensure that all students follow the rules, then you do not want to provide so many rules that it gets in the way of your planning and instruction. In many ways the teacher who is effectively managing the classroom is like a good baseball umpire: everyone knows he is there and is not afraid to enforce a rule, but he does not allow the rules to overshadow the game. In the case of the teacher, the rules do not overshadow the activities of the day. Similarly, the placement of tables, chairs, and students can help the management of the classroom. The classroom setup needs to vary depending on the instructional strategies you use. For example, you might have students in neat rows during a lecture, but they will need to be sitting in circles during a group project. More specific examples for each instructional strategy are examined later in this book.

Effective teachers also establish classroom routines that help address questions such as when to turn in homework, when to raise hands, what is expected when the bell rings at the start or end of the period, what the procedure is if someone needs to leave class, where class announcements will be posted, or what materials are needed for class (e.g., textbook, binder, pencil). By establishing clear routines, students are able to focus more on the learning and less on the logistics.

Making It Work in Your Content Area

To put some of these theories into practice, try completing three tasks.

First, consider how "daily life" in your classroom might operate; examine what you will do regarding classroom management. What routines will you use (e.g., beginning/ending class, homework collection/return, bathroom passes, etc.)? What will your policy be about late work? How will you determine seating assignments and desk arrangements?

Second, create an introductory assignment or student-interest survey that you might use to get to know more about your students. Be creative in what you ask of students; try to find out information that will assist your management of the classroom and give insight into their lives that will help you be a more effective teacher in your content area.

Third, generate a one-page handout that you might give to your students that articulates your classroom guidelines and/or expectations. These can be developed by taking "bits and pieces" from the management theories presented above. Keep in mind the following:

- Strive for a suitable level of generality.
- State your guidelines positively (say what you expect them to do) and clearly.
- Specify your expectations about students' behavior, respect for property, respect for others, academic achievement, safety, and other elements you think are important.
- List the positive and negative consequences of not adhering to your guidelines.

CHAPTER REVIEW

Effectively helping adolescents learn is a challenging task because so many variables are at work. Physically, adolescents' bodies are changing rapidly. Cognitive development during adolescence is dramatic. Thinking skills improve and develop, and adolescents begin to have more advanced cognitive abilities. Socially, adolescents begin to negotiate how they will interact with the world around them.

Learning is a "relatively permanent change, due to experience, either in behavior or in mental representations or associations" (Ormrod, 2006, p. 184). Piaget, Vygotsky, and other cognitive psychologists have investigated how our brains function, and not just our behaviors.

Gardner's multiple intelligence theory (1983, 1999) suggests that there are eight different intelligences, and that these account for a broader conception of what we think about human potential.

Motivation might best be thought of as a force, or energizer, that helps us accomplish a task. *Extrinsic motivation* is brought on by external factors that we believe will provide desirable results. *Intrinsic motivation* is motivation to engage in an activity because of the satisfaction derived from the activity itself. A useful approach for thinking about motivating students comes from a theory of intrinsic motivation called "flow."

Self-efficacy is our expectation or belief about what we can accomplish as a result of our efforts. Similar to self-efficacy is the idea of *locus of control*. If we attribute our success or failure to forces beyond our control we express an *external* locus of control. If we attribute our behavior to forces we can control we have an *internal* locus of control.

Your future classrooms will have students with different cultures, ethnicities, primary languages, economic backgrounds, religions, family structures, physical abilities, and learning abilities. *Multicultural education* is a field of study and an emerging discipline whose major aim is to create equal educational opportunities for students from diverse racial, ethnic, social-class, and cultural groups.

In your future classrooms you will have "exceptional students" who have mental retardation; learning disabilities; behavioral or emotional disorders; communication disorders; physical and/or health impairments; hearing and or vision impairments; severe or multiple disabilities; and/or, giftedness. When possible, exceptional students are kept in the general education classroom. Exceptional students are required to have an individualized educational program (IEP) or a 504 Plan.

In the decade between 1985 and 1995 the number of students in the United States with limited English proficiency grew 109%. Students learning English will tend to begin by learning social language; they learn what they use in everyday conversations. Students take much longer to negotiate their way into more formal, academic, language.

Teachers often give male students more attention and provide them with more opportunities to share during class activities. As a result, teachers may reinforce societal norms about gender-biased behaviors that actually limit girls' academic performance and enhance boys' achievements.

We have defined *technology* in reference to modern electronic resources found in classrooms that can be used for teaching and learning. Just as it is important to consider ways to include ESL students while planning lessons, it is also important to consider ways to include the digital language of our students as well as take into account our own fluency, as teachers, in that language. One of the most important things a person can learn about using technology for teaching and learning is that the focus in the classroom should not be on the technology but on how it can be used to achieve something meaningful, to promote student learning.

Classroom management is a proactive approach to helping students learn, and *discipline* is a reaction to student misbehavior. Classroom management addresses all of the aspects of the classroom that will promote student learning. Discipline specifically refers to how to effectively control students so that their behaviors maintain the sense of community in your classroom. Many conflicts and problems can be avoided if teachers try to build appropriate relationships with their students.

Establishing classroom rules and routines will also help you manage the learning environment. It is more effective to state your expectation in terms of what should be done, rather than what you do not want.

NOTES

1. Gardner admits there may be more than eight forms of intelligence, and has more recently considered additional forms such as existential or spiritual intelligence; see Gardner, 1999.
2. For more details about culturally relevant teaching, see http://www.alliance.brown.edu/tdl/tl-strategies/crt-principles.shtml (retrieved September 4, 2006).
3. For U.S. Census Bureau data about poverty, see http://www.census.gov/prod/2003pubs/p60-222.pdf (retrieved September 4, 2006).

REFERENCES

Association for Educational Communications and Technology. (1994). *Instructional technology: The definition and domains of the field*. Bloomington, IN: Author.

Bandura, A. (1994). Self-efficacy. In V. S. Ramachaudran (Ed.), *Encyclopedia of human behavior* (Vol. 4, pp. 71–81). New York: Academic Press.

Banks, J. A., & Banks, C. M. (Eds.). (1995). *Handbook of research on multicultural education*. New York: Macmillan.

Beamon, G. W. (2001). *Teaching with adolescent learning in mind*. Glenview, IL: Skylight Professional Development.

Bransford, J. D., Brown, A. L., & Cocking, R. R. (Eds.) 2000. *How people learn: Brain, mind, experience, and school.* Washington, DC: National Academy Press.

Campbell, P.B., & Steinbrueck, K. (1996). *Striving for gender equity: National programs to increase student engagement with math and science.* Washington, DC: American Association for the Advancement of Science.

CEO Forum on Education and Technology. (2001, June). *The CEO Forum school technology and readiness report: Key building blocks for student achievement in the 21st century.* Retrieved February 21, 2002, from http://www.ceoforum.org/downloads/report4.pdf.

Csikszentmihalyi, M. (1990). *Flow: The psychology of optimal experience*. New York: Harper & Row.

Csikszentmihalyi, M. (1997). *Finding flow: The psychology of engagement with everyday life*. New York: Basic Books.

Culatta, R., & Tompkins, J. (1999). *Fundamentals of special education: What every teacher needs to know.* Upper Saddle River, NJ: Prentice-Hall.

Dacey, J., Kenny, M., & Margolis, D. (2000). *Adolescent development* (3rd ed.). Carrollton, TX: Alliance.

Doda, N., George, P., & McEwin, K. (1987). 10 current truths about effective schools. *Middle School Journal, 18*(3), 3–5.

Gardner, H. (1983/2003). *Frames of mind: The theory of multiple intelligences.* New York: Basic Books.

Gardner, H. (1999). *Intelligence reframed*. New York: Basic Books.

Gardner, H. (2003, April). *20 years of multiple intelligences: Reflections and a blueprint for the future*. Keynote address made to the general membership of the American Educational Research Association, Chicago, IL.

Ginott, H. G. (1972). *Teacher and child*. New York: Macmillan.

Gorski, P. (2005). Savage unrealities: Uncovering classism in Ruby Payne's framework. Retrieved April 4, 2006 from http://www.edchange.org/publications/Savage_Unrealities.pdf.

Jones, F. H. (1987). *Positive classroom instruction*. New York: McGraw-Hill.

Ladson-Billings, G. (1995). But that's just good teaching! The case for culturally relevant pedagogy. *Theory into Practice, 34*(3), 159–165.

Lepper, M. R., & Hoddell, M. (1989). Intrinsic motivation in the classroom. In C. Ames & R. Ames (Eds.), *Research on motivation in education: Vol. 3. Goals and cognitions* (73–105). San Diego, CA: Academic Press.

Litzinger, M. E., & Osif, B. (1992). *Accommodating diverse learning styles: Designing instruction for electronic information sources.* In L. Sharito (Ed.), *What is good library instruction now?* (pp. 72–82). Ann Arbor, MI: Pierian Press.

Lortie, D. (1975). *Schoolteacher: A sociological study.* Chicago: University of Chicago Press.

Marc, P. (2001). On the horizon. *NCB University Press*, *9*(6), 1–6.

Nieto, S. (1996). *Affirming diversity: The sociopolitical context of multicultural education* (2nd ed.). White Plains, NY: Longman.

Olsen, R. W. B. (1997). Enrollment, identification, and placement of LEP students increase (again). *TESOL Matters, 7*(4), 6–7.

Oppenheimer, T. (1997). The computer delusion. *Atlantic Monthly*, *280*(1), 45–48, 50–56, 61–62.

Ormrod, J. E. (2006). *Essentials of educational psychology.* Upper Saddle River, NJ: Prentice-Hall.

Owston, R. D. (1997). The World Wide Web: A technology to enhance teaching and learning. *Educational Researcher, 26*(2), 27–33.

Padron, Y. N., Waxman, H. C., & Rivera, H. H. (2002). *Educating Hispanic students: Effective instructional practices* (Practitioner Brief #5). Retrieved February 15, 2006 from http://www.cal.org/crede/Pubs/PracBrief5.htm.

Piaget, J. (1970). Piaget's theory. In P. H. Mussen (Ed.), *Carmichael's manual of psychology* (703–732). New York: Wiley.

Pintrich, P. R., & Schunk, D. H. (2002). *Motivation in education: Theory, research, and applications* (2nd ed.). Upper Saddle River, NJ: Prentice-Hall.

Piper, M. (1994). *Reviving Ophelia: Saving the selves of adolescent girls.* New York: Ballantine Books.

Postman, N. (1995). *The end of education: Redefining the value of school*. Westminster, MD: Random House.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development and well-being. *American Psychologist*, *55*, 68–78.

Thomas, W. P., & Collier, V. P. (1997). *School effectiveness for language minority students.* Washington, DC: National Clearinghouse for Bilingual Education.

Thorne, P. (1994). *Gender play: Girls and boys in school*. Piscataway, NJ: Rutgers University Press.

Tomlinson, C. A. (2004). *How to differentiate instruction in mixed-ability classrooms* (2nd ed.). Upper Saddle River, NJ: Prentice-Hall.

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Cambridge, MA: MIT Press.

References

1 CHAPTER 1 Learning, Teaching, and Management

Association for Educational Communications and Technology. (1994). Instructional technology: The definition and domains of the field. Bloomington, IN: Author.

Bandura, A. (1994). Self-ef﾿cacy. In V. S. Ramachaudran (Ed.), Encyclopedia of human behavior (Vol. 4, pp. 71–81). New York: Academic Press.

Banks, J. A., & Banks, C. M. (Eds.). (1995). Handbook of research on multicultural education. New York: Macmillan.

Beamon, G. W. (2001). Teaching with adolescent learning in mind. Glenview, IL: Skylight Professional Development.

Bransford, J. D., Brown, A. L., & Cocking, R. R. (Eds.) 2000. How people learn: Brain, mind, experience, and school. Washington, DC: National Academy Press.

Campbell, P.B., & Steinbrueck, K. (1996). Striving for gender equity: National programs to increase student engagement with math and science. Washington, DC: American Association for the Advancement of Science.

CEO Forum on Education and Technology. (2001, June). The CEO Forum school technology and readiness report: Key building blocks for student achievement in the 21st century. Retrieved February 21, 2002, from http://www.ceoforum. org/downloads/report4.pdf.

Csikszentmihalyi, M. (1990). Flow: The psychology of optimal experience. New York: Harper & Row.

Csikszentmihalyi, M. (1997). Finding flow: The psychology of engagement with everyday life. New York: Basic Books.

Culatta, R., & Tompkins, J. (1999). Fundamentals of special education: What every teacher needs to know. Upper Saddle River, NJ: Prentice-Hall.

Dacey, J., Kenny, M., & Margolis, D. (2000). Adolescent development (3rd ed.). Carrollton, TX: Alliance.

Doda, N., George, P., & McEwin, K. (1987). 10 current truths about effective schools. Middle School Journal, 18(3), 3–5.

Gardner, H. (1983/2003). Frames of mind: The theory of multiple intelligences. New York: Basic Books.

Gardner, H. (1999). Intelligence reframed. New York: Basic Books.

Gardner, H. (2003, April). 20 years of multiple intelligences: Reflections and a blueprint for the future. Keynote address made to the general membership of the American Educational Research Association, Chicago, IL.

Ginott, H. G. (1972). Teacher and child. New York: Macmillan.

Gorski, P. (2005). Savage unrealities: Uncovering classism in Ruby Payne's framework. Retrieved April 4, 2006 from

Jones, F. H. (1987). Positive classroom instruction. New York: McGraw-Hill.

Ladson-Billings, G. (1995). But that's just good teaching! The case for culturally relevant pedagogy. Theory into Practice, 34(3), 159-165.

Lepper, M. R., & Hoddell, M. (1989). Intrinsic motivation in the classroom. In C. Ames & R. Ames (Eds.), Research on motivation in education: Vol. 3. Goals and cognitions (73-105). San Diego, CA: Academic Press.

Litzinger, M. E., & Osif, B. (1992). Accommodating diverse learning styles: Designing instruction for electronic information sources. In L. Sharito (Ed.), What is good library instruction now? (pp. 72-82). Ann Arbor, MI: Pierian Press.

Lortie, D. (1975). Schoolteacher: A sociological study. Chicago: University of Chicago Press.

Marc, P. (2001). On the horizon. NCB University Press, 9(6), 1-6.

Nieto, S. (1996). Affirming diversity: The sociopolitical context of multicultural education (2nd ed.). White Plains, NY: Longman.

Olsen, R. W. B. (1997). Enrollment, identi□cation, and placement of LEP students increase (again). TESOL Matters, 7(4), 6-7.

Oppenheimer, T. (1997). The computer delusion. Atlantic Monthly, 280(1), 45–48, 50–56, 61–62.

Ormrod, J. E. (2006). Essentials of educational psychology. Upper Saddle River, NJ: Prentice-Hall.

Owston, R. D. (1997). The World Wide Web: A technology to enhance teaching and learning. Educational Researcher, 26(2), 27–33.

Padron, Y. N., Waxman, H. C., & Rivera, H. H. (2002). Educating Hispanic students: Effective instructional practices (Practitioner Brief #5). Retrieved February 15, 2006 from http://www.cal.org/crede/Pubs/PracBrief5.htm.

Piaget, J. (1970). Piaget's theory. In P. H. Mussen (Ed.), Carmichael's manual of psychology (703–732). New York: Wiley.

Pintrich, P. R., & Schunk, D. H. (2002). Motivation in education: Theory, research, and applications (2nd ed.). Upper Saddle River, NJ: Prentice-Hall.

Piper, M. (1994). Reviving Ophelia: Saving the selves of adolescent girls. New York: Ballantine Books.

Postman, N. (1995). The end of education: Redefining the value of school. Westminster, MD: Random House.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development and well-being. American Psychologist, 55, 68–78.

Thomas, W. P., & Collier, V. P. (1997). School effectiveness for language minority students. Washington, DC: National Clearinghouse for Bilingual Education.

Thorne, P. (1994). Gender play: Girls and boys in school. Piscataway, NJ: Rutgers University Press.

Tomlinson, C. A. (2004). How to differentiate instruction in mixed-ability classrooms (2nd ed.). Upper Saddle River, NJ: Prentice-Hall.

Vygotsky, L. S. (1978). Mind in society: The development of higher psychological processes. Cambridge, MA: MIT Press.

2 CHAPTER 2 Constructing Classroom Curriculum

Apple, M. (1986). Teachers and texts. New York: Routledge and Kegan Paul.

Banks, J. A. (2003). Teaching strategies for ethnic studies (7th ed.). Boston: Allyn and Bacon.

Boyer, E. L. (1983). High school. New York: Harper and Row.

Cornbleth, C. (1991). Curriculum in context. London: Falmer.

Cuban, L. (1996). Curriculum stability and change. In P. W. Jackson (Ed.), Handbook of research on curriculum (pp. 216–247). New York: Macmillan.

Dewey, J. (1916) Democracy and education. New York: Macmillan.

Eisner, E. W. (1996). Curriculum ideologues. In P. W. Jackson (Ed.), Handbook of research on curriculum (pp. 302–326). New York: Macmillan.

Eisner, E. W. (1985). The educational imagination: On the design and evaluation of school programs (2nd ed.). New York: Macmillan.

Freire, P. (1970). The pedagogy of the oppressed. New York: Continuum.

Garcia, J., Spalding, E., & Powell, R. R. (2001). Contexts of teaching: Methods for middle and high school instruction. Upper Saddle River, NJ: Merrill/Prentice-Hall.

Gay, G. (1988). Designing relevant curricula for diverse learners. Education and Urban Society, 20, 327–340.

Jackson, P. W. (1968). Life in classrooms. New York: Holt, Rinehart, and Winston.

Hirsch, E. D. (1987) Cultural literacy. Boston: Houghton Mifﬁin.

Kliebard, H. (1987). The struggle for the American curriculum 1893–1958. London: Routledge and Kegan Paul.

National Commission on Excellence in Education. (1983). A nation at risk: The imperative for educational reform. Washington, DC: U.S. Government Printing Ofﬁce.

Of□ce of the Superintendent of Public Instruction. (2005). Washington state report card. Retrieved March 29, 2006, from http://reportcard.ospi.k12.wa.us/.

Ornstein, A., & Hunkins, F. P. (2004). Curriculum: Foundations, principles, and issues (4th ed.). Boston: Allyn and Bacon.

Posner, G. (2004). Analyzing the curriculum (3rd ed.). New York: McGraw-Hill.

Ravitch, D. (1985). The schools we deserve: Reflections on the education crisis in our lives. New York: Basic Books.

Shubert, W. H. (1986). Curriculum: Perspective, paradigm and possibility. New York: Macmillan.

Sizer, T. R. (1984). Horace's compromise: The dilemma of the American high school. Boston: Houghton Mif□in.

Thornton, S. J. (2005). Teaching social studies that matters: Curriculum for active learning. New York: Teachers College Press.

Tyler, R. W. (1949). Basic principles of curriculum and instruction. Chicago: University of Chicago Press.

United States Department of Education. (1991). America 2000: An education strategy. Washington, DC: Author.

Walker, D., & Soltis, J. (1997). Curriculum and aims (3rd ed.). New York: Teachers College Press. CITED WEBSITES

Education World's link page to all of the state standards: http://www.education-world.com/standards/state/index.shtml

"Federal Resources for Educational Excellence (FREE)" sponsored by the United States Department of Education: http://www.ed.gov/free

Professional Organizations with National Standards: Art (National Art Education Association; http://www.naea-reston.org/stdarteducation.html) English (National Council of Teachers of English; www.NCTE.org) Foreign Language (American Council on the Teaching of Foreign Languages; www.act□.org) Mathematics (National Council of Teachers of Mathematics; www.nctm.org) Music (National Association for Music Education; www.MENC.org) Physical Education (American Alliance for Health, Physical

Education, Recreation and Dance; www.aahperd.org) Science (National Science Teachers Association; www.nsta.org) Social Studies (National Council for the Social Studies; www.ncss.org) History (National Center for History in the Schools; www.sscnet.ucla. edu/nchs) Geography (National Council for Geographic Education; www.ncge. org) Economic (National Council on Economic Education; www.ncee.org) Civics (National Standards for Civics and Government; www.civiced. org)

United States Department of Education site: http://www.ed.gov/nclb This page intentionally left blank

3 CHAPTER 3 Preparing Learning Objectives and Assessing Student Learning

Armstrong, D. G., & Savage, T. V. (1994). Secondary education. New York: Macmillan.

Biehler, R. F. (1971). Psychology applied to teaching. Boston: Houghton Mifﬂin.

Black, P., & Wiliam, D. (1998a). Assessment and classroom learning. Assessment in Education, 5(1), 7–74.

Black, P., & Wiliam, D. (1998b). Inside the black box: Raising standards through classroom assessment. Phi Delta Kappan, 80(2), 139–148.

Bloom, B. S. (Ed.). (1956). Taxonomy of educational objectives: Handbook I; Cognitive domain. New York: David McKay.

Boston, C. (2002). The concept of formative assessment. Practical Assessment, Research and Evaluation, 8(9). Retrieved November 9, 2004 from http://PAREonline.net/getvn.asp?v=8&n=9.

Congress of the United States, Ofﬂce of Technology Assessment. (1992, February). Testing in American schools: Asking the right questions. Washington, DC: U.S. Government Printing Ofﬂce.

Dave, R. H. (1970). Psychomotor levels. In R. J. Armstrong (Ed.), Developing and writing behavioral objectives. Tuscon, AZ: Educational Innovators Press.

Hammond, G. (2004). Testing and evaluation. Retrieved March 2006 from http://xnet. rrc.mb.ca/glenh/outline.htm.

Hiebert, E. H., Valencia, S. W., & Afﬂerbach, P. P. (1994). Understand authentic reading assessment: Deﬂnitions and perspectives. In S. W. Valencia, E. H. Hiebert, & P. P. Afﬂerbach (Eds.), Authentic reading assessment: Practices and possibilities (pp. 6–21). Newark, DE: International Reading Association.

Kellough, R. D., & Kellough, N. G. (1999). Secondary school teaching: A guide to methods and resources; planning for competence. Upper Saddle River, NJ: Prentice-Hall.

Krathwohl, D. R., Bloom, B. S., & Masia, B. B. (1964). Taxonomy of educational objectives: Handbook 2: Affective

domain. New York: David McKay.

Mathematical Sciences Education Board and National Research Council (1993). Measuring up: Prototypes for mathematics assessment. Washington, DC: National Academy Press.

McMillan, J. H. (2004). Classroom assessment: Principles and practice for effective instruction (3rd ed.). Upper Saddle River, NJ: Allyn and Bacon.

National Council of Teachers of Mathematics (2000). The assessment principle, in Principles and Standards for School Mathematics. Retrieved December 2, 2004, from http://standards.nctm.org/document/chapter2/assess.htm.

Simpson, E. J. (1972). The classification of educational objectives in the psychomotor domain (vol. 3). Washington, DC: Gryphon House.

Slavin, R. E. (1999). Comprehensive approaches to cooperative learning. Theory into Practice, 38(2), 74-79.

Stevens, F., Lawrenz, F., & Sharp, L. (1996). User-friendly handbook for project evaluation: Science, mathematics, engineering, and technology education. Arlington, VA: National Science Foundation.

Wiggins, G. (1990). The case for authentic assessment. Practical Assessment, Research and Evaluation, 2(2). Retrieved December 6, 2004 from http://PAREonline.net/getvn.asp?v=2&n=2.

Wiggins, G. (1993). Assessment: Authenticity, context, and validity. Phi Delta Kappan, 75(3), 200-214.

Wiggins, G. (1998). Educative assessment: designing assessments to inform and improve student performance. San Francisco: Jossey-Bass.

Wiggins, G., & McTighe, J. (2000). Understanding by design. Alexandra, VA: Association for Supervision and Curriculum Development. CITED WEBSITES

Donald Clark's homepage about Bloom's taxonomy: http://www.nwlink.com/~donclark/hrd/bloom.html

RubiStar (rubric creator): http://www.rubistar4teachers.org
This page intentionally left blank

Erickson, H. L. (2002). Concept-based curriculum and instruction: Teaching beyond the facts. Thousand Oaks, CA: Corwin.

Harwood, A. M., Collins, L., & Sudzina, M. (2001). An examination of student teacher failure: Implications for teacher education programs. In J. Rainer and E. Guyton (Eds.), Teacher education yearbook (45–60). Dubuque, IA: Association of 7HDFKHU(GXFDWRUV.HQGDOO+XQW

Killion, J., & Todnem, G. (1991). A process for personal theory-building. Educational Leadership, 48(6), 14–16.

Schwab, J. J. (1973). The practical 3: Translation into curriculum. School Review, 81, 501–522.

Virginia Department of Education. (2002). Algebra I standards of learning sample scope and sequence. Richmond, VA: Virginia Department of Education

Washington State Of ce of the Superintendent of Public Instruction. (2003). Social studies frameworks: Grades 9–12. Olympia, WA: Washington State Of ce of the Superintendent of Public Instruction.

Wiggins, G., & McTighe, J. (2005). Understanding by design (2nd ed.). Alexandria, VA: Association for Supervision and Curriculum Development.

5 CHAPTER 5 Lecture and Direct Instruction

Barbetta, P. M., & Scaruppa, C. L. (1995). Looking for a way to improve your behavior analysis lectures? Try guided notes. Behavior Analyst, 18, 155–160.

Bligh, D. A. (2000). What’s the use of lectures? San Francisco: Jossey-Bass.

Brophy, J., & Good, T. (1986). Teacher behavior and student achievement. In M. Wittrock (Ed.), Handbook of research on teaching (pp. 340–370). New York: Macmillan.

Cashin, W. E. (1990). Assessing teaching effectiveness. In P. A. Seldin (Ed.), How administrators can improve teaching. San Francisco: Jossey-Bass.

Cross, K. P. (1986). A proposal to improve teaching or what taking teaching seriously should mean. AAHE Bulletin 39, 10–11.

Edwards, H., Smith, B., & Webb, G. (Eds.). (2001). Lecturing: Case studies, experience and practice. London: Kogan Page.

Engelmann, S., & Carnine, D. (1991). Theory of instruction: Principles and applications (rev. ed.). Eugene, OR: ADI Press.

Fisher, C. W., & Berliner, D. C. (Eds.). (1985). Perspectives on instructional time. New York: Longman.

Good, T. L., & Brophy, J. E. (2003). Looking in classrooms (9th ed.). Boston, MA: Allyn and Bacon.

Heward, W. L. (2005). Fast facts for faculty—guided notes: Improving the effectiveness of your lectures. Retrieved November 24, 2005 from http://telr.osu.edu/dpg/fastfact/notes.html.

Hughes, C. A., & Suritsky, S. K. (1994). Note-taking skills of university students with and without learning disabilities. Journal of Learning Disabilities, 27, 20–24.

Kame’enui, E. J., Carnine, D. W., & Dixon, R. C. (1998). Effective teaching strategies that accommodate diverse learners. In E. J. Kame’enui & D. W. Carnine (Eds.), Effective teaching strategies that accommodate diverse learners (pp. 1–17). Columbus, OH: Merrill.

Ormrod, J. E. (2006). Educational psychology: Developing learners (5th ed.). Upper Saddle River, NJ: Merrill.

Poplin, M., Wiest, D. J., & Thorson, S. (1995). Alternative instructional strategies to reductionism: Constructive, critical, multicultural, and feminine pedagogies. In W. Stainback & S. Stainback (Eds.), Controversial issues confronting special education: Divergent perspectives (2nd ed., pp. 153-165). Boston: Allyn and Bacon.

Rosenshine, B. (1987). Explicit teaching. In D. Berliner & B. Rosenshine (Eds.), Talks to teachers (75-92). New York: Random House.

Sandock, B. (2000). Enhancing learning of students with LD without compromising standards: Tips for teaching. Retrieved April 17, 2006 from http://www.ldonline.org/ld_indepth/postsecondary/facultytips.html.

Saski, J., Swicegood, P., & Carter, J. (1983). Notetaking formats for learning disabled adolescents. Learning Disability Quarterly, 6, 265-272.

Schug, M., Tarver, S., & Western, R. (2001). Direct instruction and the teaching of early reading. Wisconsin Policy Research Institute Report, 14(2), 1-27.

Wingert, D. (2001). Basic presentation skills. Retrieved April 19, 2006 from http://

Association for Direct Instruction: http://www.adihome.org

Story Telling: http://www.storyarts.org
http://www.storynet.org

National Institute for Direct Instruction:
http://www.nifdi.org

University of Minnesota's Center for Teaching and Learning Services:

Bloom, B., Englehart, M., Furst, E., Will, W., & Krathwohl, D. (1956). Taxonomy of educational objectives, Handbook 1: Cognitive domain. New York: David McKay.

Brice, L., & Johnson, L. (1999, November). Discourse as a critical pedagogical form in social studies teaching and learning. Paper presented at the annual meeting of the National Council of Social Studies, Orlando, FL.

Costa, A., & Lowery, L. (1989). Techniques for teaching and thinking. Paci□c Grove, CA: Midwest.

Dantonio, M., & Beisenherz, P. (2001). Learning to question, questioning to learn. Boston: Allyn and Bacon.

Dillon, J. T. (1988). Questioning and teaching: A manual of practice. New York: Teachers College Press.

Gall, M. (1984). Synthesis of research on teacher's questioning. Educational Leadership, 42, 40–47.

Good, T. L., & Brophy, J. E. (2000). Looking in classrooms (8th ed.). New York: Longman.

Gamoran, A., & Nystrand, M. (1992). Taking students seriously. In F. Newmann (Ed.), Student engagement and achievement in American secondary schools (40–61). New York: Teachers College Press.

Grossier, P. (1964). How to use the fine art of questioning. Englewood Cliffs, NJ: Prentice-Hall.

Henning, J. (2004). The "bow tie": A conceptual tool for opening up classroom discourse. Curriculum and Teaching Dialogue, 6, 59–67.

Hunkins, F. P. (1995). Teaching thinking through effective questioning (2nd ed.). Norwood, MA: Christopher-Gordon.

Kelly, T. (1993, April). Calling on non-volunteers in the classroom: Democratic imperative or misguided invasion? Paper presented at the annual meeting of the American Educational Research Association, Atlanta, GA.

Mills, S., Rice, C., Berliner, D., & Rousseau, E. (1980). The correspondence between teacher questions and student answers in classroom discourse. Journal of Experimental Education, 48, 194–204.

National Council for the Social Studies. (1994). Curriculum standards for social studies. Washington, DC: Author.

Newmann, F. (1992). Higher order thinking and prospects for classroom thoughtfulness. In F. Newmann (Ed.), Student engagement and achievement in American secondary schools (62–91). New York: Teachers College Press.

Newmann, F., Wehlage, G., & Lamborn, S. (1992). The signiﬁcance and sources of student engagement. In F. Newmann (Ed.), Student engagement and achievement in American secondary schools (11–39). New York: Teachers College Press.

Rowe, M. (1974). Wait time and rewards as instructional variables, their inﬂuence in language, logic, and fate control: Part one—wait time. Journal of Research in Science Teaching, 11(2), 81–94.

Rowe, M. (1986). Wait time: Slowing down may be a way of speeding up! Journal of Teacher Education, 37, 43–50.

Tanner, F. (1972). Basic drama projects. Pocatello, ID: Clark.

Tobin, K. (1986). Effects of teacher wait time on discourse characteristics in mathematics and language arts classes. American Educational Research Journal, 23(2), 191–201.

Wilen, W. (2001). Exploring myths about teacher questioning in the social studies classroom. Social Studies, 92, 26–35.

Wilen, W., Ishler, M., Hutchinson, J., & Kindsvatter, R. (2000). Dynamics of effective teaching (4th ed.). New York: Longman.

Wilen, W., & White, J. (1991). Interaction and discourse in social studies classrooms. In J. P. Shaver (Ed.), Handbook of research in social studies teaching and learning (483–495). New York: Macmillan.

Wilen, W. W. (1990). Teaching and learning through discussion. Springﬁeld, IL: Charles C Thomas.

Wilen, W. W. (1991). Questioning skills for teachers (3rd ed.). Washington, DC: National Education Association.

White, J. J. (1990). Involving different social and

cultural groups in discussion. In W. W. Wilen (Ed.), Teaching and learning through discussion (pp. 147–157). Spring[illegible]eld, IL: Charles C Thomas. This page intentionally left blank

Ehrenberg, S. D. (1981). Concept learning: How to make it happen in the classroom. Educational Leadership, 39(1), 36–43.

National Council for Social Studies (1994). Curriculum standards for social studies: Expectations of excellence. Washington, DC: Author.

Parker, W. C. (1987). Teaching thinking: The pervasive approach. Journal of Teacher Education, 38(3), 50–56.

Parker, W. C. (2005). Social studies in elementary education (12th ed.). New York: Prentice-Hall.

Parker, W. C., & Perez, S. (1987). Beyond the rattle of empty wagons. Social Education, 52(3), 164–166.

Taba, H. (1967). Teacher's handbook. Palo Alto, CA: Addison-Wesley.

Vygotsky, L. S. (1978). Mind in society: The development of higher psychological processes. Cambridge, MA: MIT Press.
CITED WEBSITE

Wikipedia: http://www.wikipedia.org This page intentionally left blank

8 CHAPTER 8 Cooperative Learning

Aronson, E., Blaney, N., Stephen, C., Sikes, J., & Snapp, M. (1978). The jigsaw classroom. Beverly Hills, CA: Sage.

Blosser, P. E. (1992). Using cooperative learning in science education. ERIC Bulletin, SE 053 432. Retrieved February 3, 2006 from http://www.stemworks.org/Bulletins/SEB92-1.html.

Cohen, E. G. (1994). Designing groupwork: Strategies for the heterogeneous classroom. New York: Teachers College Press.

Cohen, E., & Lotan, R. (1997). Working for equity in heterogeneous classrooms: Sociological theory in practice. New York: Teachers College Press.

Doda, N., George, P., & McEwin, K. (1987). 10 current truths about effective schools. Middle School Journal, 18(3), 3–5.

Heller, P., Keith, R., & Anderson, S. (1992). Teaching problem solving through cooperative grouping. Part 1: Group versus individual problem solving. American Journal of Physics, 60(7), 627–636.

InTime (2004). Chapter one: Cooperative learning overview. Retrieved February 8, 2006 from http://www.intime.uni.edu/coop_learning/ch1/types.htm.

Johnson, D. W., & Johnson, R. T. (1989, December–1990, January). Social skills for successful group work. Educational Leadership, 47(4), 29–33.

Johnson, D. W., & Johnson, R. T. (1991). What cooperative learning has to offer the gifted. Cooperative Learning, 11(3), 24–27.

Johnson, D. W., & Johnson, R. T. (1999). Learning together and alone: Cooperative, competitive, and individualistic learning. Boston: Allyn and Bacon.

Johnson, D. W., & Johnson, R. T. (2004). Assessing students in groups: Promoting group responsibility and individual accountability. Thousand Oaks, CA: Corwin.

Johnson, D. W., Johnson, R. T., & Holubec, E. J. (1994). The nuts and bolts of cooperative learning. Edina, MN: Interaction.

Johnson, D. W., Johnson, R. T., & Holubec, E. (1998). Cooperation in the classroom. Boston: Allyn and Bacon.

Johnson, D. W., Johnson, R. T., & Stanne, M. B. (2000, May). Cooperative learning methods: A meta-analysis. Retrieved April 6, 2006 from http://clcrc.com/pages/cl=methods.html.

Kagan, S. (1992). Cooperative learning. San Juan Capistrano, CA: Resources for Teachers.

Kohn, A. (1993). Punished by rewards: The trouble with gold stars, incentive plans, A's, praise, and other bribes. Boston, MA: Houghton Mif□in.

Kulik, J. A., & Kulik, C. L. C. (1987). Effects of ability grouping on student achievement. Equity and Excellence, 23, 22–30.

Kulik, J. A., & Kulik, C. L. C. (1991). Research on ability grouping: Historical and contemporary perspectives. Storrs, CT: University of Connecticut, National Research Center on the Gifted and Talented. (ERIC Document Reproduction Service No. ED 350 777).

Marzano, R. J., Pickering, D. J., & Pollock, J. E. (2001). Classroom instruction that works: Research-based strategies for increasing student achievement. Alexandria, VA: Association for Supervision and Curriculum Development.

Newmann, F. M., & Thompson, J. A. (1987, September). Effects of cooperative learning on achievement in secondary schools: A summary of research. (ERIC Document Reproduction Service No. ED 288 853).

Parker, W. C. (2005). Social studies in elementary education. (12th ed.). Upper Saddle River, NJ: Merrill/Prentice-Hall.

Sharan, Y., & Sharan, S. (1989, December–1990, January). Group investigation expands cooperative learning. Educational Leadership, 47(4), 17–19.

Slavin, R. (1993). Ability grouping in the middle grades: Achievement effects and alternatives. Elementary School Journal, 93, 535–552.

Slavin, R. E. (1983). Cooperative learning. New York: Longman.

Slavin, R. E. (1988, June). Student team learning: An overview and practical guide. Washington, DC: National Education Association.

Slavin, R. E. (1989, December–1990, January). Research on cooperative learning: Consensus and controversy. Educational Leadership, 47(4), 52–54.

Slavin, R. E. (1995). A model of effective instruction. Educational Forum, 59, 166–176.

Wiggins, G. (1990). The case for authentic assessment. Practical Assessment, Research and Evaluation, 2(2). Retrieved December 6, 2004 from http://PAREonline.net/getvn.asp?v=2&n=2.

Zarnowski, M. (2003). History makers: A questioning approach to reading and writing biographies. Portsmouth, NH: Heinemann.

9 CHAPTER 9 Simulations, Role-Play, and Dramatization

%UDQVIRUG⍰-⍰'⍰%URZQ⍰$⍰/⍰
&RFNLQJ⍰5⍰5⍰(GV⍰How people learn: Brain, mind, experience, and school. Washington, DC: National Academy Press.

Farris, P. J., & Parke, J. (1993). To be or not to be: What students think about drama. Clearinghouse, 66(4), 231–234.

Fisher, J., & Vander Laan, S. (2002). Improving approaches to multicultural education: Teaching empathy through role-playing. Multicultural Education, 9, 25–27.

Gardner, H. (1999).

Greenblat, C. S. (1987). Designing games and simulations. Beverly Hills, CA: Sage.

Johnson, A. P. (1998). How to use creative dramatics in the classroom. Childhood Education, 75(1), 2–82.

-RQHV⍰.⍰Simulations: A handbook for teachers and trainers⍰UG⍰HG⍰(DVW⍰ Brunswick, NJ: Nichols.

-R\FH⍰%⍰:HLO⍰0⍰ ⍰&DOKRXQ⍰(⍰Models of teaching (6th ed.). Boston: Allyn and Bacon.

Newmann, F., Bryk, A., & Nagaoka, J. (2001). Authentic intellectual work and stanGDUGL]HG⍰WHVWV⍰&RQÁLFWV⍰RU⍰FRH[LVWHQFH"⍰Chicago: Consortium on Chicago School Research. Retrieved March 1, 2006 from http://www.consortium-chicago.org/.

Smith, J., Lee, V., & Newmann, F. (2001). Instruction and achievement in Chicago elementary schools. Chicago: Consortium on Chicago School Research. Retrieved March 1, 2006 from http://www.consortium-chicago.org/.

Shaftel, F., & Shaftel, G. (1967). Role playing of social values.⍰(QJOHZRRG⍰&OLIIV⍰1-⍰ Prentice-Hall.

Wiggins, G., & McTighe, J. (2005). Understanding by design (2nd ed.). Alexandra, VA: Association for Supervision and Curriculum Development. This page intentionally left blank

10 CHAPTER 10 Classroom Discussion and Debate

$OYHUPDQQ'(2·%ULHQ'*
'LOORQ'5:KDWWHDFKHUVGRZKHQWKH\ say they are having discussions of content area reading assignments: A qualitative analysis. Reading Research Quarterly, 25(4), 296–322.

%HOODFNS$.OLHEDUG+0+\PDQ5 7
6PLWK(/-UThe language of the classroom. New York: Teachers College Press.

Bridges, D. (1979). Education, democracy and discussion.:LQGVRU(QJODQG1DWLRQDO
)RXQGDWLRQIRU(GXFDWLRQDO5HVHDUFK

Bridges, D. (1987). Discussion and questioning. Questioning Exchange, 1, 34–37.

Cazden, C. (1988). Classroom discourse. Portsmouth, NH: Heinemann.

&RKHQ(*Designing groupwork: Strategies for heterogeneous classrooms, New York: Teachers College Press.

Dillon, J. T. (1984). Research on questioning and discussion. Educational Leadership, 41(3), 51–56.

Dillon, J. T. (1994). Using discussion in the classroom. Philadelphia: Open University Press.

(QJOH6 2FKRD$ Education for democratic citizenship: Decision making in the social studies. New York: Teachers College Press.

Gall, M. D. (1985). Discussion methods of teaching. In T. Husen & T. N. PostlethZDLWH(GV International encyclopedia of education (Vol. 3, pp. 1423–1427). Oxford: Pergamon.

Gall, M. D., & Gall, J. P. (1990). Outcomes of the discussion method. In W. W. Wilen (G Teaching and learning through discussion: The theory, research and practice of the discussion method SS²6SULQJÀHOG,/&KDUOHV&7KRPDV

*RRG&9(G Dictionary of education. New York: McGraw-Hill.

Goodlad, J. I. (1984). A place called school. New York: McGraw-Hill.

Gorman, A. H. (1969). Teachers and learners: The interactive process. Boston: Allyn and Bacon.

*URVV5(=HOHQ\/'Educating citizens for democracy: Curriculum and instruction in secondary social studies. New York: Oxford University Press.

Haroutunian-Gordon, S. (1991). Turning the soul: Teaching through conversation in the high school. Chicago: University of Chicago Press.

Hess, D. (2000). Developing strong voters through democratic deliberation. Social Education, 64(5), 193–196.

Hoetker, J., & Ahlbrand, W. P., Jr. (1969). The persistence of recitation. American Educational Research Journal, 6, 145–167.

+RRYHU.+The professional teacher's handbook: A guide for improving instruction in today's middle and secondary schools (2nd ed.). Boston: Allyn and Bacon.

Hunkins, F. P. (1995). Teaching thinking through effective questioning (2nd ed.). Boston: Christopher-Gorden.

Johnston, J., Anderman, L., Milne, L., & Harris, D. (1994). Improving civic discourse in the classroom: Taking the measure of Channel One. Research Report 4. Ann Arbor, MI: Institute for Social Research, University of Michigan.

Johnson, D. W., & Johnson, R. T. (1988). Critical thinking through structured controversy. Educational Leadership, 45(8), 58–64.

/DUVRQ%(&RPSDULQJ IDFH WR IDFH GLVFXVVLRQ DQG HOHFWURQLF GLVFXVVLRQ A case study from high school social studies. Theory and Research in Social Education, 31(3), 348–366.

0DLO ' - 7KH ÃVKERZO 'HVLJQ IRU GLVFXVVLRQ Today's Education, 57, 27–29.

Miller, S. (1992). Creating change: Towards a dialogic pedagogy. (Report Series 2.18). Albany, NY: National Research Center on Literature Teaching and LearnLQJ(5,&'RFXPHQW5HSURGXFWLRQ6HUYLFH1R('

Oliver, D. W., & Newmann, F. M. 1972). Taking a stand: A guide to clear discussion of public issues

Parker, W. C. (2001). Classroom discussion: Models for leading seminars and deliberations. Social Education, 65(2), 111-115.

3DUNHU:&0F'DQLHO-(
9DOHQFLD6:+HOSLQJVWXGHQWVWKLQNDERXW
public issues: Instruction versus prompting, Social Education, 55(1), 41-44.

3ULOHV00$ 7KH ÃVKERZO GLVFXVVLRQ$ VWUDWHJ\ IRU
ODUJH KRQRUV FODVVHV English Journal, 82(6), 49-50.

6FKZDE - -

6WRGROVN\6)HUJXVRQ7
:LPSHOEHUJ.7KHUHFLWDWLRQSHUVLVWV EXW ZKDW
does it look like? Journal of Curriculum Studies, 13(2), 121-130.

Swift, J. N., & Gooding, C. T. (1983). Interaction of wait time feedback and questioning instruction on middle school science teaching. Journal of Research in Science Teaching, 20(8), 721-730.

Tharp, R. G., & Gallimore, R. (1988). Rousing minds to life: Teaching, learning, and schooling in social context. Cambridge: Cambridge University Press.

Wilen, W. W., & White, J. J. (1991). Interaction and discourse in social studies
classURRPV,Q-3KDYHU(GHandbook of research on social studies teaching and learning (pp. 483-495). New York: Macmillan. CITED WEBSITES

Blackboard Inc.: http://www.blackboard.com

DiscusWare: http://www.discusware.com This page intentionally left blank

11 CHAPTER 11 Student-Directed Investigation

Boekaerts, M. (1997). Self-regulated learning: a new concept embraced by researchers, policy makers, educators, teachers, and students. Learning and Instruction, 7, 161–186.

Bransford, J. D., Brown, A. L., & Cocking, R. R. (Eds.). (2000). How people learn: Brain, mind, experience, and school. Washington, DC: National Academy Press.

Brooks, J. G., & Brooks, M. B. (1993). The case for constructivist classrooms. Alexandria, VA: Association for Supervision and Curriculum Development.

Brown, A. L. (1997). Transforming schools into communities of thinking and learning about serious matters. American Psychologist, 54(4), 399–413.

Brown, H. (2004). Walking into the unknown: Inquiry-based learning transforms the English classroom. English Journal, 94(2), 43.

Brown, J. S., Collins, A., & Duguid, P. (1989). Situated cognition and the culture of learning. Educational Researcher, 18(1), 32–42.

Bruce, B. C., & Davidson, J. (1996). An inquiry model for literacy across the curriculum. Journal of Curriculum Studies, 28, 281–300.

Cognition and Technology Group at Vanderbilt (1990). Anchored instruction and its relationship to situated cognition. Educational Researcher, 19(5), 2–10.

Collins, A., Brown, J. S., & Newman, S. E. (1994). Cognitive apprenticeship. In L. B. Resnick (Ed.), Knowing, learning, and instruction (pp. 453–494). Hillsdale, NJ: Lawrence Erlbaum Associates.

Duffy, T., & Jonassen, D. (Eds.). (1992). Constructivism and the technology of instruction: A conversation. Hillsdale, NJ: Lawrence Erlbaum Associates.

Exline, J. (2004). Inquiry-based learning. Retrieved February 1, 2006 from http://www.thirteen.org/edonline/concept2class/inquiry/.

Fosnot, C. T. (1989). Inquiring teachers, enquiring

learners. New York: Teachers College Press.

Gallagher, S., Sher, B., Stepien, W., & Workman, D. (1995). Implementing problembased learning in science classrooms. School Science and Mathematics, 95(3), 136–146.

Harris, K., Marcus, R., McLaren, K., & Fey, J. (2001). Curriculum materials supporting problem-based teaching. School Science and Mathematics, 101(6), 310–318.

McGrath, D. (2003). Developing a community of learners: What will it look like and how will it work? Learning and Leading with Technology, 30(7), 42–45.

Newmann, F., Bryk, A., & Nagaoka, J. (2001). Authentic intellectual work and standardized tests: Conflicts or coexistence? Chicago: Consortium on Chicago School Research. Retrieved February 15, 2006 from http://www.consortiumchicago.org/publications/p0001.html.

Pithers, R. T., & Soden, R. (2000). Critical thinking in education: A review. Educational Research, 42(3), 237–249.

Saye, J. W., & Brush, T. (1999). Student engagement with social issues in a multimedia-supported learning environment. Theory and Research in Social Education, 27(4), 472–504.

Scheurman, G. (1998). From behaviorist to constructivist teaching. Social Education, 62, 6–9.

Scheurman, G., & Newmann, F. M. (1998). Authentic intellectual work in social studies: Putting performance before pedagogy. Social Education, 4(3), 150–153.

Skinner, B. F. (1968). The technology of teaching. New York: Appelton.

Struthers, K. (2005). Assessing the credibility of online sources. Retrieved February 1, 2006 from http://leo.stcloudstate.edu/research/credibility1.html.

Tomlinson, C. (2003). Fulfilling the promise of differentiated classrooms: Strategies and tools for responsive teaching. Alexandria, VA: Association for Supervision and Curriculum Development.

West, S. A. (1992). Problem-based learning—A viable addition for secondary school science. School Science Review, 73(265), 47–55.

Wilke, R., & Straits, W. (2005). Practical advice for teaching inquiry-based science process skills in the biological sciences. American Biology Teacher, 67(9), 534–540.

Wisker, G., & Brown, S. (Eds.). (1996). Enabling student learning. London: Kogan Page/Staff and Educational Development Association.

Appendix B State Standards and Assessments

Cummins, J. (1979). Cognitive/academic language proﬁciency, linguistic interdependence, the optimum age question and some other matters. Working Papers on Bilingualism, 19, 121–129.

Freire, P. (1978). Pedagogy of the oppressed. New York: Continuum. This page intentionally left blank